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Review

Review of Film Cooling Techniques for Aerospace Vehicles

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Abstract: Film cooling, a vital method for controlling surface temperatures in components subjected to intense heat, strives to enhance efficiency through innovative technological advancements. Over the last several decades, considerable advancements have been made in film cooling technologies for applications such as liquid rocket engines, combustion chambers, nozzle sections, gas turbine components, and hypersonic vehicles, all of which operate under extreme temperatures. This review presents an in-depth investigation of film cooling, its applications, and its key mechanisms and performance characteristics. The review also explores design optimization for combustion chamber components and examines the role of gaseous film cooling in nozzle systems, supported by experimental and numerical validation. Gas turbine cooling relies on integrated methods, including internal and external cooling, material selection, and coolant treatment to prevent overheating. Notably, the cross-flow jet in blade cooling improves heat transfer and reduces thermal fatigue. Film cooling is an indispensable technique for addressing the challenges of high-speed and hypersonic flight, aided by cutting-edge injection methods and advanced transpiration coolants. Special attention is given to factors influencing film cooling performance, as well as state-of-the-art developments in the field. The challenges related to film cooling are reviewed and presented, along with the difficulties in resolving them. Suggestions for addressing these problems in future research are also provided.

Keywords: film cooling; hypersonic vehicle; cooling effectiveness; liquid coolant; slot configurations

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1. Introduction

Advanced aerospace applications such as hypersonic vehicles, reusable launch vehicles, modern gas turbine engines, and long-duration missions require thermal protection systems (TPSs) that are lightweight, durable, and reusable [1]. These TPSs are crucial for managing extreme heat during atmospheric re-entry, hypersonic flight, rocket engine operation, and in environments near high-temperature propulsion systems [2,3]. Thermal protection systems are employed in aerospace components exposed to extreme temperatures, preventing structural damage, ensuring vehicle integrity, and protecting payloads and crew by managing the intense heat generated from aerodynamic friction, combustion, or radiation [1]. Instances that require high-temperature environments include conditions such as spacecraft re-entry temperatures surpassing 1650 °C (3000 °F), the extreme heat encountered during hypersonic flight [2], and turbine inlet temperatures exposed to hot combustion gasses. Traditional thermal protection techniques, such as Ultra-High-Temperature Ceramics (UHTCs), additive manufacturing, active cooling systems, and adaptive materials [3], are employed in various aerospace applications. Other thermal protection system (TPS) methods, including regenerative cooling, film cooling, transpiration

cooling, and combined cooling strategies, are employed in these applications. The applications of traditional TPS include re-entry capsules such as the Apollo Command Module, SpaceX Dragon, the Space Shuttle's silica-based tiles designed to withstand re-entry heat [1], the nose and wing leading edges of the Space Shuttle [3], and thermal management systems used in satellites and space probes [2]. The limitations associated with traditional TPS can be explained as follows: Ablative materials are not reusable, have significant weight, and are unsuitable for reusable vehicles; ceramic tiles are delicate, easily damaged, and require extensive maintenance [1]. Reinforced Carbon-Carbon (RCC) Composites are costly, weighty, and challenging to produce [3]. Insulating Blankets are only effective in lower-temperature scenarios and perform poorly under extreme heat conditions [2]. Film cooling is an indispensable thermal protection technique used in aerospace vehicles, particularly in components subjected to extreme temperatures. It is commonly used in rocket engines, gas turbines, and hypersonic vehicles. The advantages of the film cooling technique over other thermal protection methods can be summarized as follows: film cooling significantly reduces surface temperatures by creating a thermal barrier between the structure and extreme heat. Compared to ablative materials, film cooling is reusable and does not break down over time [1]. It eliminates the need for heavy thermal protection materials, boosts fuel efficiency, raises payload capacity, and enables precise targeting of coolant to high-heat areas, thus enhancing thermal protection efficiency [4]. Furthermore, it safeguards critical components, such as turbine blades, guide vanes, combustion chambers, and hypersonic vehicles against thermal stress and oxidation, thus prolonging their operational lifespan [5]. These advantages over other cooling techniques make it the most suitable method for applications in combustion chambers, turbine components, and hypersonic systems. The design and cooling of the combustion chamber are crucial for maintaining engine performance and durability [6]. Similarly, the turbine blade cooling system must be capable of cooling all areas exposed to high-temperature gas flow [7]. Film cooling is essential for the thermal management of hypersonic vehicles, protecting critical components from extreme temperatures while ensuring the vehicle's durability and aerodynamic efficiency [8]. A combination of film and impingement cooling is employed for the thermal management of turbine blades. These techniques are used to cool the leading edge, while turbulent rib cooling with serpentine channels is applied to the middle section, and cooling of the trailing edge is achieved through pin-fin technology [7].

The combustion chambers of engines such as the SSME, F-1, J-2, RS-27, Vulcain 2, RD-171, and RD-180 incorporate the film cooling method [9]. Several open-cycle rocket engines, such as the F1 [10] and J2 [11] engines from the United States, utilize turbine exhaust gas (TEG) for film cooling in the nozzle. In rocket engines, the film coolant in the combustion chamber creates a protective layer that limits the movement of combustion products toward the walls, thereby decreasing the rate at which the walls oxidize [9]. This cooling technique is also applied in Japan's upgraded LE5 engine and the Vulcain-2 engine by EADS Astrium [12]. Investigations have shown that the studies on film cooling originated in the late 1800s, beginning with the work of Reynolds and Lamb in fluid mechanics. Reynolds [9] investigated the formation and movement of vortex rings, an area strongly linked to the representation of film cooling jets. The applications of various cooling techniques in actual aerospace engines are summarized in Table 1. It can be seen that the RS-25 engine mainly employed regenerative cooling, utilizing liquid hydrogen, along with film cooling, to manage the intense thermal stresses encountered during operation. This strategy guarantees the engine's reliability and efficiency in demanding spaceflight conditions. Similarly, the F-1 engine utilized regenerative cooling with RP-1 fuel, enhanced by film and ablative cooling, to manage severe thermal stresses. This comprehensive cooling strategy was vital for the engine's performance in the Saturn V rocket. The J-2 engine also

used regenerative cooling with liquid hydrogen, supplemented by film and dump cooling, to address thermal stresses. This integrated cooling system was critical for maintaining engine cooling effectiveness in the top stages of the Saturn V and Saturn IB rockets. The RS-27 engine used regenerative cooling with RP-1 fuel, augmented by film cooling and ablative protection, to withstand extreme thermal loads. This cooling strategy proved indispensable for first-stage performance in both the Delta II and Delta III rockets. The Vulcain 2 engine used regenerative cooling with liquid hydrogen as the primary method, supplemented by film and dump cooling to mitigate extreme thermal stresses. This design is critical for its function in the first stage of the Ariane 5 rocket. The RD-171 engine utilized regenerative cooling with RP-1 fuel, supported by film and turbopump exhaust gas cooling to manage extreme thermal challenges. This comprehensive cooling method is critical for maintaining engine cooling effectiveness in the first stage of the Zenit rocket. Finally, the RD-180 engine relied on regenerative cooling with RP-1 fuel as the primary method, supplemented by film and turbopump exhaust gas cooling to manage thermal stresses. This design guarantees its performance in the first stage of the Atlas V rocket.

Table 1. Film cooling techniques and their applications in actual engines.

Engine	Cooling Technique	Coolant/ Propellant	Vehicle
RS-25 [9]	Regenerative	Liquid-fuel cryogenic	Space Shuttle and Space Launch System (SLS)
F-1 [10]	Regenerative (Primary cooling method)	RP-1	Saturn V rocket
	Film and ablative cooling (Secondary cooling methods)	Ablative materials	
RS-27 [9]	Regenerative (Primary cooling method)	RP-1	Delta II and Delta III rockets
	Film and ablative cooling (Secondary cooling methods)	Ablative materials	
J-2 [11]	Regenerative (Primary cooling method)	Liquid hydrogen (LH2)	Saturn V and Saturn IB rockets
	Film and dump cooling (Secondary cooling method)	Liquid hydrogen (LH2)	
Vulcain 2 [9]	Regenerative (Primary cooling method)	Liquid hydrogen (LH2)	Ariane 5 rocket
	Film and dump cooling (Secondary cooling methods)	Liquid hydrogen (LH2)	
LE5 [12]	Regenerative	Liquid hydrogen (LH2)	H-I and H-II rockets
RD-180, RD-171 [9]	Regenerative (Primary cooling method)	RP-1	Atlas V rocket and Zenit rocket
	Film and turbopump gas cooling (Secondary cooling methods)	Exhaust gas	

Scientific investigations on film cooling applications in turbine elements and rocket combustion chambers began in the 1950s [9]. Since then, researchers have conducted numerous experimental studies and developed multiple models to evaluate the effectiveness of

film cooling techniques. Current research focuses on optimizing cooling designs, reducing coolant usage, and integrating film cooling with other thermal management techniques. The thermal protection techniques employed in aerospace vehicles depend on material designs to either absorb, disperse, or reflect heat to protect the vehicle's structure and components from extreme temperature conditions [1]. This review highlights advancements in film cooling technologies, with a particular emphasis on gaseous and liquid film cooling, as well as alternative approaches for combustion chambers, gas turbine blades, and hypersonic vehicles. It highlights cutting-edge research in design and materials while underscoring the critical role of computational and experimental validations in evaluating cooling efficiency. These innovations aim to enhance the durability, performance, and safety of aerospace systems exposed to extreme thermal stresses.

This review aims to assess the effectiveness of film cooling in applications such as combustion chambers, turbine blades, and hypersonic vehicles while identifying knowledge gaps, highlighting the significance of the current research, critically reviewing previous studies, and laying the groundwork for future investigations.

1.1. Advances in Gaseous Film Cooling Research

Gaseous film cooling, widely used in aerospace applications, is vital for boosting the thermal protection of high-temperature components in rocket engines and hypersonic vehicles. It creates a cool gas barrier that protects surfaces from intense heat, enhancing durability and performance. This technology is crucial for maintaining the safety and efficiency of aerospace systems in extreme thermal conditions. Over the years, research on film cooling has primarily concentrated on gaseous film cooling in gas turbine airfoils due to its widespread use in aviation and power generation. Likewise, numerous studies have also explored cooling strategies for combustion chambers used in rocket applications, addressing the challenges posed by extreme heat and pressure in space propulsion systems. Both areas of study aim to enhance thermal protection and performance in high-temperature environments.

Progress in film cooling technology, particularly gaseous film cooling, has focused on a comprehensive review of its applications in aerospace vehicles over the last seven decades. This includes an in-depth analysis of hypersonic aerodynamics, with special attention to critical thermal protection methods used in hypersonic flights [2]. Studies have also focused on protecting components exposed to aerothermal stresses using well-designed thermal protection techniques [13]. Progress has also been made in areas such as the functional capacities of hypersonic systems, including re-entry vehicles, space propulsion, and interceptor systems when the gaseous film cooling technique is employed [14]. Recent studies have focused on how passive and active strategies enhance film cooling effectiveness [15]. Gaseous film cooling studies have been reviewed, providing an in-depth analysis of the methodologies and the various factors influencing film cooling performance [5]. An experimental analysis has been conducted to determine how the positions of injected fluid flow on blades with different curvatures affect film cooling effectiveness [16]. Various film cooling techniques, including internal and external cooling configurations, have been adopted to investigate the effects of turbine blade design factors on overall cooling effectiveness [17]. Experimental investigations were conducted to analyze the influence of blowing ratios on slot film-cooling effectiveness of a turbine blade shroud, involving blade rotation and various hole and slot configurations [18]. Similarly, the efficiency of the gaseous film cooling technique used for film-cooling a gas turbine rotor casing has been experimentally and numerically investigated, focusing on coolant ejection techniques [19]. The surface temperature distribution and gaseous cooling efficiency within a gas turbine combustion chamber have been studied to determine the absolute temperature values [20]. Also, a modified

model of the reaction chamber, similar to that used in the Vulcain 2 engine, was employed to predict gaseous film cooling effectiveness at varying reaction chamber pressures and blowing ratios across different Mach regimes [21]. In another experimental study coupled with a theoretical investigation, a convergent-divergent nozzle was employed to study the heat transfer effects, the role of the nozzle throat in cooling film stability, and the appropriate methods for effectively analyzing the impacts of boundary and mixing layers [22]. An experimental study on gaseous film cooling efficiency in a Laval nozzle operating under varying Mach regimes was conducted to investigate the role of an enhanced mainstream turbulence effects [23]. In another study, a central body for the propulsion exhaust used in a turbofan engine was investigated both experimentally and numerically to examine the influence of flow behavior and infrared radiation distribution of the propulsion exhaust system, featuring film cooling with and without a central body for various geometries [24]. A comprehensive outline of the difficulties faced in high-temperature gas flow regions of gas turbine engines has been presented, including recent approaches and a preliminary design guide for future research advancements [25]. Studies involving theoretical and numerical methods were conducted to examine the gaseous film cooling efficiency of a 2D flat plate in a hypersonic environment [26]. In this study, a calculation method was derived theoretically and used numerically to determine the effects of shockwaves, coolant flow inclination, and mainstream velocity on film cooling effectiveness.

This review outlines key unresolved challenges and research gaps in applying gaseous film cooling to aerospace vehicles:

(1) Although interdependencies currently exist to determine the efficiency of gaseous film cooling used in aerospace applications under various injection parameters, a significant decrease in overall heat flux directly corresponds to an increase in the heat transfer coefficient. However, this does not always guarantee improved film cooling effectiveness due to the intricate interaction between heat transfer, fluid dynamics, and film stability;
(2) The effect of thrust nozzle side loads during transient operation is not fully understood, and the mechanisms driving this phenomenon have not yet been identified or explained;
(3) Success in addressing major challenges for air-breathing hypersonic vehicles, such as thermal robustness, aerodynamic stability, and manufacturing complexities, depends heavily on specific vehicle requirements. These challenges make selecting and implementing a reliable thermal management technique difficult;
(4) Thermal protection materials with the required heat resistance, oxidation stability, and hypersonic-temperature durability are scarce. Thus, additional studies are necessary to establish accurate models for cutting-edge thermal protection systems;
(5) The reciprocal interplay between the discrete jet and the free stream flow is not fully understood. Therefore, a deeper understanding of the mechanisms governing fluid interaction dynamics of jets in crossflow and thin-layer cooling performance attributes is essential for developing advanced film cooling models for real-world applications;
(6) The review found that most research on film cooling technology primarily focuses on analyzing cooling efficiency, and other factors, such as the heat transfer coefficient, the effects of side loads, and total thermal performance efficiency, are often overlooked. Advanced research incorporating these factors will provide a more detailed understanding of state-of-the-art thermal barrier optimization strategies;
(7) The rationale for choosing shaped holes over cylindrical holes, including using a unified thermal management approach to minimizing flow performance degradation in PTS schemes, has not been thoroughly explained. Further studies and analyses are needed to evaluate their respective advantages;

(8) Due to the complexity of fluid flow, few CFD models can accurately simulate the effects of coolant jet detachment. Additionally, there is currently limited data to comprehensively analyze the role of factors such as primary flow turbulence intensity, surface curvature, and hole geometry. Further studies are needed to develop advanced models for accurately analyzing the mechanisms of coolant jet detachment and predicting film cooling effectiveness in the design of turbine engine components.

Available experimental and numerical studies were reviewed, and it was determined that these studies lack sufficient data to resolve these research gaps. Below, some of the methods and film-cooling techniques reviewed are presented. Anderson [2] provided a comprehensive overview of hypersonic aerodynamics, emphasizing the critical thermal protection methods required for designing hypersonic vehicles. Consequently, a well-designed TPS should offer dependable protection against aerothermal stresses, avoiding excessive weight increases or compromising the vehicle's structural integrity [13]. As the Mach number increases, hypersonic vehicles encounter significant challenges such as intense heat flux, severe thermal gradients, high stagnation pressures, and reactive plasma from gas ionization, which hastens the oxidation of materials [3,14,27]. Glass [3] addressed these issues by presenting aerothermal effects and strategies for heat control techniques required to manage intense heating. This included an outline of TPS strategies specifically suited for vehicles utilizing rocket propulsion and those relying on air-breathing engines. Similarly, Chen [27] conducted a numerical study to model the interaction between materials and environmental conditions in a hypersonic thermal protection system. The outline of this investigation includes the design of an integrated simulation model, computational analysis of emitted radiation, simulation of SiC oxidation and nitridation reactions, and modeling of rate-dependent surface oxidation variables.

A review of how passive and active strategies enhance film cooling effectiveness [15] highlights current breakthroughs in sophisticated film cooling methods. It suggests that thermal protection for hot-section components remains one of the most significant complexities in advancing propulsion engine technology. Bogard and Thole [5] evaluated the performance of the gaseous film cooling technique by assessing the effects of upstream turbulence, surface curvature, and hole geometry on film cooling effectiveness. Although the review provided insights into the magnitude of these effects, the available data are currently limited. Ito et al. [16] experimentally investigated the effects of surface curvature on the film cooling performance of a wall-aligned jet by determining the actual position of the injected coolant relative to the wall surface. They observed that the surface curvature near the coolant holes significantly contributes to cooling effectiveness. Town et al. [17] employed a simplified single-dimensional model to analyze film cooling effectiveness in a turbine rotor blade, incorporating internal and external cooling designs. They utilized a coolant plenum to provide internal impingement cooling and external film cooling for the leading edge, and impingement cooling was applied to the inner surface of the leading edge. The trailing edge was film-cooled using a coolant plenum that supplied impingement cooling. This model has proven useful for selecting the appropriate design approaches for turbine rotor blade thermal protection systems (TPS). Tamunobere and Acharya [18] experimentally investigated the film cooling of a blade shroud in a low-velocity rotating turbine test rig by simultaneously injecting coolant through a freestream slot and inclined discrete holes. They evaluated the film cooling performance separately for coolant injected through the freestream slot, inclined discrete holes, and a combined technique. Based on blowing ratios, the results showed that the combined cooling technique provided significantly higher cooling effectiveness compared to the individual methods across all blowing ratios studied. Similarly, Collins et al. [19] used both experimental and numerical methods to investigate the cooling efficiency of a turbine rotor casing. In their transient 3D

simulation model, they observed that the coolant discharge holes on the turbine rotor casing produced a complex flow pattern. This phenomenon significantly reduced film effectiveness values at the pressure-side edge of the casing due to the impingement of the mainstream flow. Feist et al. [20] used thermographic phosphors to analyze film cooling effectiveness, assessing the surface thermal conditions over a targeted section within a combustion chamber and on the peripheral components of a gas turbine. Their goal was to determine the feasibility of this technique, which employs a ceramic host matrix with a lanthanide ion dopant. The measured values were consistent with the experimental results obtained using a similar method.

Arnold et al. [21] investigated film cooling effectiveness in a nozzle section of a miniature rocket combustion chamber using wall-mounted and surface-mounted thermal probes. They utilized temperature distributions at different Mach regimes to evaluate film cooling efficiency in the presence of high-velocity hot gas. Similarly, a numerical investigation was conducted to analyze the effects of blowing ratio and crosswise corrugation geometry on adiabatic film cooling effectiveness and heat transfer coefficient [28]. The 3D numerical results confirmed a significant advantage of the crosswise corrugation geometry, and the adiabatic film cooling effectiveness was compared with an effusion-based cooling film. It was also observed that the crosswise corrugation geometry resulted in reduced adiabatic film cooling effectiveness and an increased heat transfer coefficient compared to the model with a flat surface. Stoll and Straub [22] conducted experimental and theoretical investigations into heat transfer effects and the stability of cooling films in a convergent-divergent nozzle. Their analysis involved injecting pressurized gas into high-temperature air at the nozzle inlet. It was determined that the throat section had no significant effects on film stability when assessing film-cooling effectiveness. Lebedev et al. [23] investigated the role of optimized mainstream turbulence intensity on the kinetic and thermal flow in a Laval nozzle. They applied the Kutateladze-Leont'ev theory of cooling gas layers to extrapolate the empirical findings for nozzle flows and complex flow phenomena. It was observed that the optimized mainstream turbulence intensity model yielded a significant reduction in cooling effectiveness in both subsonic and supersonic regimes. Shan et al. [24] conducted experimental and numerical investigations to examine the influence of film cooling hole geometry on boundary layer temperature and infrared radiation suppression in a full-scale propulsion exhaust system. The numerical model was analyzed using bidirectional ray propagation techniques and a selective wavelength model, which were validated by the experimental results. Their experimental findings indicated a reduction in infrared radiation suppression within the propulsion exhaust system, and the numerical results demonstrated that the mass flow rate into the central body depended on the diameter of the film-cooling holes and the surface area of the fixed mounting points at the inlet. Consequently, reducing the diameter of the film-cooling holes led to a decrease in infrared radiation suppression.

Bunker [25] presented an overview of the challenges associated with high-temperature gas flow regions in gas turbine engines, analyzing recent approaches and proposing a preliminary design guide for future research advancements. However, the review did not recommend specific solutions or rank the discussed challenges in order of importance. Zhou et al. [26] proposed that film cooling, considered an effective active cooling technique, could be utilized in hypersonic vehicle technologies. Since the surface of a hypersonic vehicle is simplified as a flat plate rather than a non-streamlined body, the researchers investigated the cooling properties of a flat plate in a two-dimensional setup under hypersonic conditions, employing theoretical analysis and numerical techniques. They developed a method to calculate thermal performance in hypersonic environments, and the numerical results demonstrate strong predictive accuracy. Gaseous film cooling is a promising technique for application in nuclear thermal rockets and high-thrust liquid

chemical propulsion systems [9]. However, the constraints of modern gas turbines are largely determined by the capabilities of current cooling technologies used for hot-section components [25]. Over time, research on film cooling has mainly concentrated on gaseous film cooling for gas turbine blades; more recent studies have expanded to include the film cooling of rocket combustion chambers [9]. The current review has demonstrated that most studies on gaseous film cooling technology have primarily focused on experimental investigations, with limited numerical efforts being made. Additionally, there is a lack of CFD research utilizing the smooth-particle hydrodynamic (SPH) method for thermal protection system (TPS) designs and validations. Due to the SPH method's advantages over other CFD techniques—especially its ability to handle large deformations and complex geometries—further studies using the SPH are strongly encouraged. Similarly, additional studies are required to develop TPS models that fully capture the intricate interactions between heat transfer, fluid dynamics, and cooling film stability.

1.2. Advances in Liquid Film Cooling Research

The liquid film cooling mechanism differs from gaseous film cooling due to the liquid's higher heat capacity and the latent heat released during phase change, which significantly enhances cooling efficiency. In liquid film cooling, it has been noted that the vaporized coolant does not quickly mix with the mainstream gas flow; instead, it remains as a protective vapor layer next to the surface, extending a significant distance in the flow direction from where the liquid layer ends [9].

Studies on liquid film cooling have witnessed significant progress over the years, and a summary of such advancements is presented in this section. This film cooling technique has been employed in aerospace applications, such as the internal film cooling of combustion chambers in rocket engines with known chamber pressures and oxidant-to-fuel mixture ratios [29]. This study utilized controlled percentages of coolants like water, ethyl alcohol, and liquid ammonia. An experimental study examined the effects of surface film cooling performance in a rocket engine using cryogenic–storable propellants [30]. The study included a detailed evaluation of available quantitative heat transfer techniques. The influence of liquid coolant injector geometry on cooling efficiency has been investigated through experimental studies [31]. The study quantified the effect of blowing ratio on cooling film coverage and thermal uniformity downstream of the injection point. An experimental study of the film cooling process on a rigid surface subjected to a high-momentum turbulent flow was conducted [32]. The study focused on identifying factors that prevent liquid film detachment and minimize the loss of unevaporated liquid when simple radial-hole injectors are used. The effects of parameters such as mass, momentum, and flow characteristics like heat transfer were numerically investigated using a liquid film cooling technique in the rocket combustion chamber [33]. The study employed a coupled approach to solve the individual governing equations and model the interaction with the freestream gas. This review highlights the following unresolved problems and research gaps in liquid film cooling for aerospace vehicles:

(1) Surface film cooling performance in a rocket engine utilizing a combination of cryogenic–storable propellants resulted in abrupt propellant depletion. Although the researchers suggested several techniques to prevent this phenomenon, these techniques were not validated for real-world applications. Therefore, more experimental and numerical studies should be conducted to understand this phenomenon;
(2) A key challenge in advancing film cooling hole designs is the cost-effective manufacturing of conceptual designs while maintaining mechanical properties under hot combustion gasses. Thus, more research should focus on developing advanced manufacturing techniques for these designs.

A summary of the recent advances in the field, analyzing how researchers and designers addressed key challenges associated with liquid film injection techniques applicable to aerospace vehicles, is presented as follows:

Morrell [29] conducted experimental studies on liquid film cooling using a vertical rectangular slot injector and a tangential-oriented coolant passage type with slots oriented at 45° to the axis. However, the tangentially oriented slot injection technique did not enhance cooling effectiveness compared to the vertical slot injection technique. Similarly, Kesselring et al. [30] conducted experiments using tangential coolant injectors in a nickel calorimetric chamber. They pinpointed the crucial design factors and emphasized their respective significance. Data simplification techniques were created to analyze the thermal energy uptake of the vaporized film in short-term high-temperature experiments. However, calculations of the anticipated heat flux without transpiration led them to conclude that the liquid film evaporated abruptly. Shine et al. [31] conducted studies on both experimental and computational methods on cylindrical coolant holes with tangential and multi-angle configurations. They observed that incorporating a compound angle did not enhance cooling effectiveness compared to tangential injection. Additionally, the tangential injector showed a longer liquid film length than the compound angle injector. A key observation is that, in most studies, coolant holes are angled tangentially to the hot gas flow [9].

An experimental study investigated film cooling on a rigid boundary surface under high-momentum turbulent flow. This led to the development of a technique to calculate the evaporation rate and surface temperature for a uniform inert thermal barrier [32]. A study involving the effects of entrainment and reacting films in rocket combustion chambers was conducted to ascertain and optimize the thermal management and performance potential of swirling in a liquid film [34]. The findings revealed reduced entrainment and manageable wall thermal temperature. Zhang et al. [33] used a coupled approach to numerically investigate the effects of flow characteristics and coolant properties in film cooling for a rocket combustion chamber. The turbulence effects of the liquid coolant were simulated and analyzed using a simplified van Driest model. Subsequently, the mainstream gas flow was simulated using a heat transfer model. The in-depth investigation of how these parameters affect the cooled length of the liquid film showed numerical predictions consistent with experimental results.

Experimental, numerical, analytical, and theoretical studies of liquid film cooling in rocket thrust chambers are presented in Tables 2–7. Table 2 focuses on film viability analysis, Table 3 on applications in the Nozzle Section, Table 4 on Coolant Types, Table 5 on Heat Transfer and Evaporation Analysis, Table 6 on Flow Phenomena and Transpiration Analysis, and Table 7 on Performance Analysis, flow modeling, and related topics. Liquid film cooling is indispensable in aerospace for the thermal protection of components such as turbines and rocket engines. Creating a cooling layer reduces heat exposure and enhances the durability, performance, and safety of systems operating under extreme thermal stress. It should be noted that, due to proprietary restrictions in spacefaring nations and companies, detailed information on film cooling in liquid-based rocket propulsion systems is scarcely available in published research [9].

The current review has demonstrated the limited studies on liquid film cooling compared to gaseous film cooling for applications in aerospace technology. Additionally, it is noted that very few investigations on liquid film cooling have focused on applications in hypersonic conditions. Due to the advantages of the SPH method over other CFD approaches, further studies on liquid film cooling for hypersonic applications using the SPH method are necessary to understand the intricate flow interactions between a hypersonic vehicle's surface and the extreme gas temperature environment. Although most studies have focused on the effects of blowing ratios and coolant injection geometry, a

few numerical investigations have explored these factors in film cooling for hypersonic vehicle surfaces.

Table 2. Film viability analysis in liquid film cooling for rocket thrust chambers: experimental investigations.

Source	Research Approach	Area of Investigation, Injector Types
Shine and Shri Nidhi [9] Warner and Emmons [35] Knuth [32]	Experimental	Feasibility of films, Radial injectors
Warner and Emmons [35]	Experimental	Feasibility of films, Circumferential slots
Morrell [29] Shine and Shri Nidhi [9] Stechman et al. [36] Kinney [37] Crocco [38] Abramson [39] Marek and Tacina [40] Spalding [41] Shine and Shri Nidhi [9] Hombsch and Olivier [42] Shine et al. [43]	Experimental	Feasibility of different coolants, Vertical and Tangential slots
Shine et al. [44]	Experimental	Feasibility of films, Straight-cylindrical holes
Shine et al. [43]	Experimental	Feasibility of films, Compound-cylindrical holes
Warner and Emmons [35]	Experimental	Feasibility of H_2 as a Coolant, Dual slot radial injector
Arrington et al. [45]	Experimental	Feasibility of different coolants, Standard Conical Nozzle Cooling

Table 3. Experimental and numerical investigations of liquid film cooling in the nozzle section of rocket thrust chambers.

Source	Research Approach	Area of Investigation, Injector Types
Abramson [39] Stoll and Straub [22] Arrington et al. [45] Marek and Tacina [40] Spalding [41] Shine and Shri Nidhi [9] Hombsch and Olivier [42]	Experimental	Cooling of nozzles, Tangential slots
Volkmann et al. [46] Shine and Shri Nidhi [9] Stoll and Straub [22]	Experimental	Cooling of the nozzle throat

Table 3. *Cont.*

Source	Research Approach	Area of Investigation, Injector Types
Martelli et al. [47] Arrington et al. [45]	Experimental	Cooling of the bell nozzle throat,
Martelli et al. [47] Matesanz et al. [48]	Numerical	Film injection in dual bell nozzle, Convergent-divergent nozzles
Badinand and Fransson [49]	Numerical	Film-cooled nozzle, Radiative heat transfer
Wang and Guidos [50] Cikanek [51] Watanabe et al. [52]	Numerical	Film-cooled nozzle extension, Side -load physics

Table 4. Coolant types in liquid film cooling for rocket thrust chambers: experimental and numerical investigations.

Source	Research Approach	Area of Investigation
Stechman et al. [36] Arrington et al. [45]	Experimental	Propellants as Coolants
Cook and Quentmeyer [53] Kirchberger et al. [54]	Experimental	Hydrocarbon as a Coolant
Kirchberger et al. [55]	Experimental	Kerosene as a Coolant
Zhang et al. [33]	Numerical	Coolant loss is estimated through vapor diffusion

Table 5. Experimental and analytical investigations of liquid film cooling in rocket thrust chambers, including heat transfer and evaporation analysis.

Source	Research Approach	Area of Investigation
Kanda et al. [56]	Experimental	Influence of the external shockwave, Thermal conductance parameter
Shine and Shri Nidhi [9]	Analytical	Evaluation of heat transfer coefficient
Shine and Shri Nidhi [9]	Analytical	Transpiration, radiative heat transfer, and turbulence from the free-stream
Crocco [38]	Analytical	Evaporation process in liquid films
Knuth [32]	Analytical	Techniques for estimating the evaporation rate of liquid films

Table 6. Experimental and analytical investigations of liquid film cooling in rocket thrust chambers, including flow phenomena and transpiration analysis.

Source	Research Approach	Area of Investigation, Injector Types
Warner and Emmons [35] Knuth [32] Shine and Shri Nidhi [9]	Experimental	Stability of liquid films, Radial injectors
Spalding [41]	Experimental	Coolant jet on flow separation
Gater et al. [57]	Analytical	Model including film instability and transpiration effects
Stechman et al. [36]	Analytical	Introduction 'flow instability efficiency correction factor'
Yu et al. [34]	Analytical	Swirling of the Liquid Film

Table 7. Experimental, theoretical, and numerical investigations of liquid film cooling in rocket thrust chambers, including performance analysis, flow modeling, and related aspects.

Source	Research Approach	Area of Investigation, Injector Types
Kinney [37]	Experimental	Study of performance, Film visualization, and Porous and jet-type analysis
Kesselring et al. [30] Shine et al. [43] Arnold et al. [21] Morrell [29]	Experimental	Development of analytical model, Tangential injector
Spalding [41] Shine and Shri Nidhi [9]	Experimental and Theoretical	Correlation for effectiveness, Tangential injector
Shine et al. [31]	Experimental and Numerical	Tangential and compound cylindrical
Di Matteo et al. [58] Shine and Shri Nidhi [9]	Numerical	Film wall jet in combustion chambers
Shembharkar and Pai [59]	Numerical	Couette flow model
Wang and Luong [60]	Numerical	Regeneratively cooled engine

1.3. Other Film Cooling Techniques for Aerospace Components

Film cooling in rocket engines is typically used alongside other cooling techniques, most commonly regenerative cooling. Over the years, various cooling methods, including film, radiation, transpiration, heat sink, ablative, and dump cooling, have been improved to decrease the demand for regenerative cooling and minimize propellant usage [9]. Regenerative cooling is the typical cooling method used in nearly all contemporary primary propulsion systems, boosters, and engines designed for the final phase of propulsion [61].

This section reviews the main progress on transpiration and regenerative cooling techniques. The transpiration cooling technique has been used to analyze the impact of turbulence on cooling efficiency over a flat plate subjected to hypersonic flow, focusing on coolant injection factors. Experiments on laminar transpiration cooling examined film cooling behavior in hypersonic flow under external conditions. The study measured the coolant gas concentration near the wall and the heat flux reduction downstream of a transpiring injector in hypersonic laminar flow, utilizing a High-Density Tunnel at Mach 7 [62]. In these experiments, six test cases utilized nitrogen and helium as coolant gasses with varying blowing ratios, and heat flux reduction was achieved across all test conditions. Even at the lowest blowing ratio, film coverage exceeding 15% was recorded over three injector lengths [9]. Adriano [63] conducted a computational study to examine the effects of coolant injection through porous media in a hypersonic turbulent surface layer environment. The analysis examined factors such as blowing ratio and the influence of pore diameter on cooling effectiveness. The simulation exposed a flat plate to hypersonic gas flow generated by a porous injector geometry. The results identify key factors contributing to flow instability in porous boundary layer transpiration cooling. Based on these findings, parameter optimization is essential to ensure effective boundary layer cooling under hypersonic flow instability.

Film cooling, a crucial technology for enhancing the durability of liquid rocket engines, works by placing a protective barrier of coolant fluid over the surface that needs protection from the high-temperature gas flow [9]. This technique is depicted in Figure 1. It is observed that coolant in rocket engine combustion chambers is injected through slots or holes and flows along high-temperature walls. The film cooling process encompasses three key effects, as shown in Figure 2: (a) thermal energy is transferred from the wall to the fluid via convection; (b) a film is generated by the coolant gas, which acts as an insulator on the wall surface, creating a thermal barrier against the hot cross-flow; and (c) the wall is protected from oxidation due to the thermal barrier between the wall and the free-stream oxygen. This allows the wall material to operate at elevated temperatures, enhancing radiative cooling and reducing recombination-induced heating [62]. Film cooling is achieved through two methods: internal cooling (used in combustion chambers), as shown in Figure 1, and external cooling (employed in modern turbines and the surfaces of hypersonic vehicles), as depicted in Figure 2.

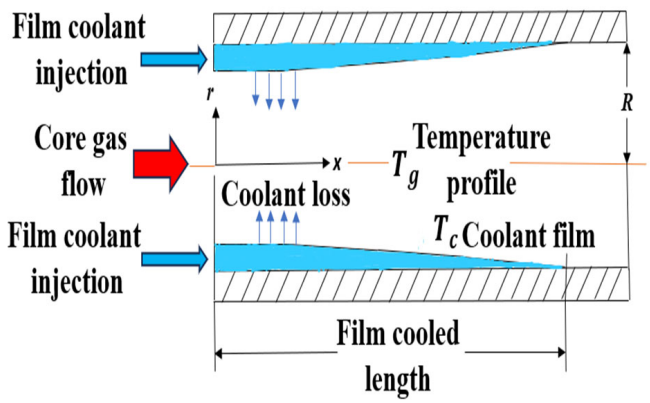

Figure 1. Diagram of the physical system—internal cooling (reproduced from [9]).

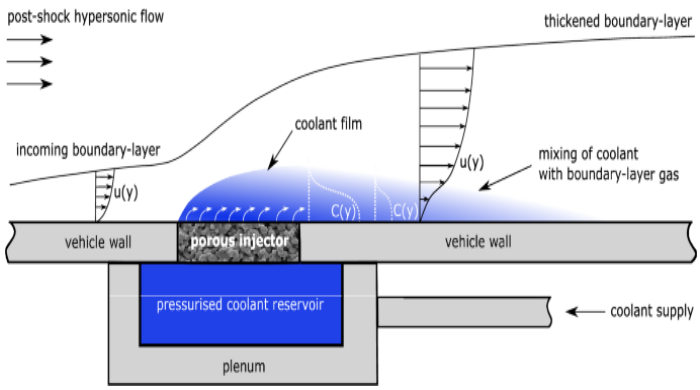

Figure 2. Experiments on laminar transpiration in hypersonic flow—external cooling (adapted from [62]).

1.4. Validation of Coolant Injector Hole Configuration

Research on improving the efficiency of film cooling technology aims to optimize cooling performance and reduce coolant consumption by validating various techniques through computational and experimental methods. Model validations are critical for optimizing existing models to achieve enhanced designs that improve engine efficiency and reduce operational costs. Studies validating thermal protection system models are reviewed and summarized below. Techniques for the reduction or elimination of Counter Rotating Vortices (CRV), also known as kidney vortices, caused by secondary flow in the coolant jet on a flat surface, were developed by NEKOMIMI cooling technology [64]. The formation of CRV leads to a phenomenon called flow separation, which in turn results in a reduction in film cooling effectiveness. The NEKOMIMI technology maintains the cooling stream close to the flat wall, ensuring uniform distribution along the surface. The first anti-kidney vortices model, developed by NEKOMIMI technology utilizing the integration of double holes in a single arrangement, proved challenging to manufacture even with state-of-the-art manufacturing techniques. Hence, this technology was enhanced using a collection of geometric variables, resulting in an arrangement that was easy to fabricate while maintaining the original technology's effectiveness. Coolant injector hole arrangements, such as cylindrical and shaped geometries designed to counter kidney vortices in the coolant jet, have been used to numerically study the effects of adiabatic and coupled heat transfer phenomena in a gas turbine blade [65]. A 3D injection configuration was numerically studied to develop and validate the model using experimental data from individual cooling jets [66]. This configuration was subsequently applied to simulate a film-cooled nozzle guide vane.

The current review identifies the research problems in validating the coolant injector hole configurations used in aerospace applications. These problems are summarized and presented as follows:

(1) Fabricating advanced anti-kidney vortex configurations remains a challenge. Hence, further research is needed into manufacturing techniques that can simplify the production of these advanced configurations;

(2) It is nearly impossible to account accurately for the secondary effects caused by the complex interaction of fluid dynamics and thermal energy exchange. These effects include secondary thermal convection due to the intricate nature of the coolant film thermal performance in gas turbine blade design. This performance is typically evaluated using a standard approach for assessing adiabatic film-cooling efficiency, which also depends on the surrounding thermal performance environment.

Various methods have been used to develop and validate advanced coolant hole geometries, and some of these techniques are presented below. Kusterer et al. [64] based their findings on a numerical analysis of geometric parameters. They created new configurations that exhibited very high adiabatic film cooling effectiveness at a known blowing ratio under ambient airflow conditions similar to those of a wind tunnel test rig. Additionally, they performed experimental studies to compare the standard NEKOMIMI configuration with a more advanced version. Although the advanced configuration did exhibit enhanced film cooling performance, the degree of improvement was less than anticipated based on the numerical analysis. Bohn et al. [65] conducted a numerical study of duct flow involving cooling fluid injected through various hole configurations, including non-shaped and shaped exits. They demonstrated that using conjugate heat transfer analysis accounts for the impact of heat transfer on the velocity field within the cooling film. Their findings revealed that the fan-shaped design is up to three times as effective as the cylindrical hole design. Kusterer et al. [64] conducted a numerical parametric study that utilized various advanced configurations and examined their performance under high-temperature gas flow environments. They systematically varied parameters to achieve further optimization for maximum film cooling effectiveness. Innovations in cooling techniques frequently investigate new designs, such as shaped holes, trenches, and compound angle injections, to enhance cooling efficiency and adapt to increasing turbine inlet temperatures. Dahlander et al. [66] developed an injection model using fully resolved 3D simulations and validated it against experimental data. Their results demonstrated promising agreement in predicting the heat transfer characteristics of the injection configuration.

Computational and experimental validations involve developing advanced computational fluid dynamics (CFD) models and experimental setups to more accurately predict and evaluate film cooling performance across various operational scenarios.

2. Recent Progress and Innovations in Film Cooling Technology

This section reviews emerging trends, innovative designs, and methodological breakthroughs in film cooling technology. The review focuses on the dynamics of cooling slot flows [67] and the primary determinants of cooling effectiveness [44,68], including CFD, a vital tool for evaluating and improving film cooling performance in aerospace applications. These advancements collectively aim to enhance the efficiency and reliability of thermal management systems in high-temperature environments. Key investigations into the challenges of cooling slot flow dynamics are reviewed and presented in Section 2.1. Similarly, an outline of specific research advancements and applications of the effects of coolant injection hole orientation, injector types, number of cooling slots, investigation methods, freestream turbulence, and shockwaves on film cooling efficiency is presented in Section 2.2. Section 2.3 reviews the latest trends and advancements in film cooling technology for aerospace applications. These trends and advances encompass experimental, numerical, theoretical, and analytical approaches, along with related technical considerations, presented in chronological order. Finally, Section 2.4 reviews the various CFD methods applied in film cooling research.

2.1. Cooling Slot Flow Dynamics

The effect of varying injection angles on the effectiveness of film cooling techniques has been a major focus of research. Significant progress in cooling slot configurations has provided insights into the flow dynamics of these advanced technologies. The key advancements are reviewed and presented below in chronological order. Progress on cooling slots for thermal management in aerospace applications began in the early 1950s. The annular slot injection technique was used to experimentally investigate film cooling

effectiveness in a liquid-ammonia and liquid-oxygen propulsion rocket nozzle [39]. It is worth noting that a decrease in the efficiency of the uncooled rocket engine was observed. This phenomenon was significant enough to render the application of the annular injection technique undesirable for this case. Another experimental study utilized water as the film coolant, injected at a single axial point [37]. The experimental results revealed similarities in the thermal transfer rates obtained with 2-inch and 4-inch diameter even-surfaced tubes compared to those obtained in an earlier investigation using a 4-inch diameter uneven-surfaced tube. In 1964, a study was conducted to experimentally analyze the influence of various flow parameters on the liquid film-cooled length of different dimensions, injected radially through peripheral slots in a cylindrical rocket engine combustor, using high-temperature mainstream air at a known flow rate [35]. In 1975, an experimental study was conducted to analyze the effects of turbulence intensity on film cooling by evaluating the film-cooled surface temperatures downstream of turbulence-inducing generators in a rectangular duct [40]. The study compared the quantified streamwise turbulence intensity and the calculated turbulent transport coefficient, revealing insufficient results. Similarly, the results showed that the streamwise turbulence coefficient decreased compared to those obtained with directional mixing intensity.

More than three decades later, experimental studies were conducted to investigate the influence of the wall thermal profile and film cooling efficiency, including a reduction in heat transfer rate at the wall caused by tangentially induced film cooling [69]. The investigation utilized gaseous hydrogen as a coolant, introduced through a tangential slot injection technique near the outer surface of the injector. In the 2010s, experimental studies were conducted in a small-scale, high-pressure experimental combustor using a combination of cryogenic propellants injected axially toward the surface plate [67]. The experiments identified critical flow factors influencing film cooling effectiveness, the thermal efficiency of the axially injected film, and the reduction in thermal load at the wall. Another experimental investigation examined the effect of film coolant injection geometry on cooling efficiency, utilizing gaseous and liquid cooling media [31]. The experiments employed hot air as the mainstream flow and the gaseous nitrogen and water as film coolants, which were injected through straight and compound-angle injection configurations using a cylindrical test rig to simulate a thrust chamber. Film cooling effectiveness was examined using a flat plate subjected to laminar and turbulent boundary-layer flow fields [42]. Coolant was introduced to the plate's surface through various blowing configurations to determine the cooling efficiency. A study numerically examined the flow distribution related to cylindrical cooling jets oriented in tangential and circumferential directions [43]. The injection configurations of the cooling jets inside a circular pipe were examined.

The challenges encountered in investigating coolant slot flow dynamics for aerospace vehicles are identified and presented as follows:

(1) Numerous investigations involving slot flow dynamics have primarily focused on experimental studies. Future investigations on this topic should employ numerical methods, particularly the SPH (Smoothed Particle Hydrodynamics) method, due to its ability to handle complex fluid interactions. This approach will provide deeper insights into the flow dynamics of various injection configurations;
(2) The ring-shaped or annular slot injection technique used in propulsion rocket nozzle cooling effectiveness analysis is considerably unsuitable. Additional experimental and numerical investigations are necessary to gain insights into the ring-shaped technique and to develop advanced, suitable annular slot injection configurations for this application.

Key investigations addressing the challenges of cooling slot configurations are summarized as follows: Abramson [39] conducted internal thermal management experiments

on the propulsion nozzle of an ammonia-oxygen propulsion system using an annular slot injection technique. Although film cooling was successfully achieved across the entire nozzle using coolants such as water and anhydrous ammonia, the efficiency was significantly reduced. Kinney [37] utilized porous and jet-type injectors, featuring holes angled at 25° to the axis. The results showed no substantial difference between the two types of coolant injectors. Warner and Emmons [35] introduced coolant radially through peripheral slots in a gaseous hydrogen-fueled thrust chamber of a rocket engine. Their results revealed that a two-slot injection technique significantly reduced the amount of coolant required to film-cool a specific surface length compared to a single-slot approach. Marek and Tacina [40] utilized tangential slots to introduce coolant air into a rectangular test rig, investigating the impact of varying ambient turbulence on film cooling effectiveness. Their results demonstrated a significant decrease in film cooling effectiveness as the ambient turbulence intensity increased. Similarly, experiments by Arnold et al. [69] demonstrated that injector design significantly affects wall temperatures, while enhanced-pressure combustion conditions produce more pronounced circumferential temperature variations. Arnold et al. [67] conducted further experiments in a smaller propulsion chamber, implementing tangential hydrogen film cooling. The study assessed tangential slot injection under different film-cooling parameters. Their findings revealed that increasing the slot Reynolds number directly improved film-cooling effectiveness. Research conducted by Shine et al. [31] demonstrated that the injector configurations significantly influenced film cooling performance, particularly at elevated blowing ratios. The experiment by Hombsch and Olivier [42] further validated the effectiveness of film cooling techniques in laminar flow conditions compared to turbulent flow conditions. Their findings revealed that film cooling is significantly more effective in laminar flows and that tangential injection outperforms angled slot injection. Shine et al. [43] investigated the flow distribution through circular coolant passages tilted in both tangential and azimuthal directions within a circular pipe. Their findings showed that optimal configurations for coolant injectors could yield maximum cooling performance. A notable increase in heat transfer coefficients was observed downstream of the injection for all injector configurations.

This review reveals that some studies focused on slot film cooling, while others examined discrete coolant holes. Significant efforts have been made in recent decades to improve discrete-jet film cooling. The optimization of discrete-hole designs, including advanced geometries and cooling slot configurations, shows promising results in minimizing hot spots and improving thermal management. Additionally, advanced simulation techniques and experimental methods provided valuable insights into the dynamics of coolant flow, leading to more accurate predictions of cooling performance. Combining these optimized parameters with modern materials and manufacturing techniques opens new possibilities for enhancing the efficiency and longevity of high-temperature systems.

2.2. Determinants of Film Cooling Effectiveness

Film cooling effectiveness depends on geometric parameters, and a few studies have specifically focused on how the injection angle impacts thermal protection performance [9]. These parameters include flow conditions of the coolant and mainstream (such as turbulence in the mainstream [40], coolant Mach number [70], mainstream Mach number [71,72], and blowing ratio [44,68]), geometric parameters (like injector geometry [31,69,73], injector orientation [43], and surface curvature [74]), as well as other elements such as heat capacity [75]. The potential coupling between these factors suggests that synergistic effects could impact film cooling performance. As a result, a wide range of operating conditions may exist, making it inherently challenging to predict cooling efficiency [9]. Table 8 provides an overview of studies on various parameters affecting film cooling effectiveness.

Table 8. Elements influencing the performance of film cooling (data from [9]).

Coolant/Freestream Flow State	Geometric Variables	Additional Considerations
Turbulence in freestream, *Tu* [40]	Injector design [31,69,75]	External Shock Wave [9,56]
Coolant Mach No [70]	Injector orientation [43]	Swirl in the Mainstream [76]
Mainstream Mach No [56,71,72] Blowing ratio, M [44,68]	Surface curvature [74]	Heat capacity, c_c [75]

Geometric orientation, such as introducing a sliding block in the nozzle injector design [70], can enhance cooling effectiveness. The orientation of the sliding block in the nozzle injection design, tapered at a 10-degree angle, is shown in Figure 3. Similarly, the orientation of the mainstream and secondary flows [71] strongly impacts cooling effectiveness via shock wave formation. Figure 4 shows the locations of the mainstream flow, secondary flows, and the surface exposed to high-temperature gas.

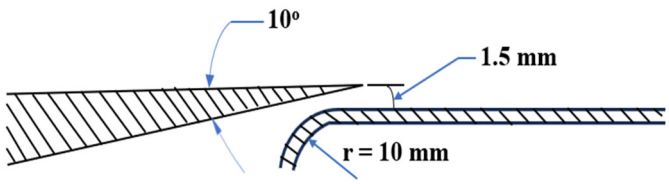

Figure 3. Shape and structure of the injection nozzle (reproduced with permission from [70], Elsevier, 2025).

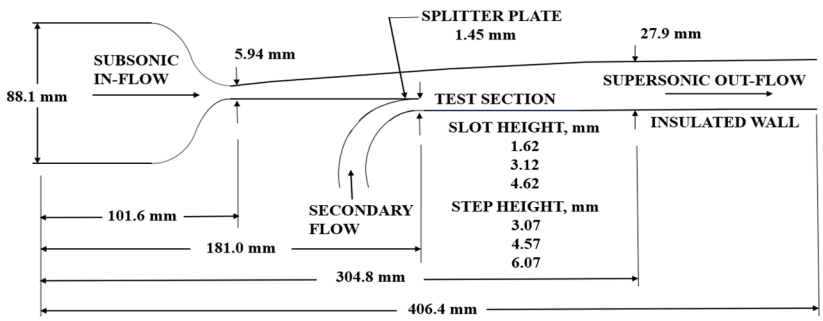

Figure 4. Configuration and fluid flow details (reproduced with permission from [71], Elsevier, 2025).

An in-depth review of the main progress on the factors determining film cooling efficiency is presented chronologically, providing insights into the underlying principles and their applications. In 1975, an experimental study validated the inverse relationship between film-cooling effectiveness and mainstream turbulence intensity [40]. In this experiment, the cooling stream was introduced at the ambient flow temperature at the inlet, representing a known percentage of the total airflow. It was observed that the cooling efficiency decreased as the mainstream turbulence intensity increased, thereby validating the hypothesis. In the 1990s, a numerical study was conducted to examine the influence of mainstream Mach number [71] on film cooling effectiveness by injecting high-temperature secondary airflow at supersonic conditions. The injected airflow was introduced through a backward-configured slot, and the numerical findings, compared with experimental data, indicated effective cooling performance. Another experimental investigation evaluated the influence of coolant Mach number and other parameters, such as velocity and mass

transfer rate, on cooling performance [70]. The study utilized helium as the coolant, and the adiabatic surface temperature was directly measured. The findings revealed a significant increase in film cooling performance corresponding to a rise in the coolant injection rate. Similarly, the role of the mainstream Mach number on film cooling effectiveness was investigated using experimental and analytical methods in both subsonic and supersonic regimes [72]. The results concluded that film cooling is effective in the subsonic regime due to low-temperature occurrence, while the opposite is evident in supersonic regimes.

An experimental and numerical study was conducted to examine the influence of geometric parameters on the effectiveness of the film cooling technique in a rocket propulsion system under supersonic flow conditions [73]. The study also examined various turbulence simulation approaches and compressibility modifications using a boundary-layer computational model for validation. The results concluded that the explicit models examined were inappropriate for this simulation. Meanwhile, in the 2000s, factors including the blowing ratio, mainstream Mach and Reynolds numbers, and geometric parameters such as blowing geometry were investigated using experimental and numerical approaches in a laminar flow at supersonic speeds [68]. The findings revealed similarities with those obtained in an earlier study on laminar supersonic flow regimes. Another experimental investigation involving a tangential slot injection configuration was performed in a small-scale combustion chamber [69]. The findings revealed notable differences in wall temperatures due to the coolant injection configuration, which led to a uniform distribution of cooling effectiveness downstream of the coolant injection locations. The projection of cooling effectiveness, based on the effects of blowing ratios and geometric factors, was examined using experimental and numerical approaches [44]. It was observed that increased blowing ratios had negligible effects on coolant mixing and high vortex formation. In another experimental investigation, gaseous and liquid coolants were used to study the influence of geometric parameters, such as injector geometry, on film cooling efficiency, film cooling coverage length, and cooling layer homogeneity [31]. In the liquid film cooling investigation, the recorded findings revealed that injecting coolants through a compound-angled geometry diminished far-field cooling efficiency and reduced film cooling coverage compared to those achieved with a tangential injection configuration. The orientation of the cooling jet is also a determining factor in investigating film cooling effectiveness across various flow regimes [43]. Coolant jets tilted in the tangential and circumferential directions, injected through cylindrical holes, were employed to study the flow distribution inside a cylindrical pipe. The observed results were compared with proprietary experimental data, and the findings suggested that the most efficient coolant injector designs promote a reduction in coolant usage. In 2025, surface curvatures, such as flat and curved surfaces, were employed to examine the influence of parameters such as Mach number at the cooling gas intake and blowing ratio under supersonic flow regimes, with experimental validation of film cooling efficiency [74]. The findings concluded that the nature of the geometric surface has no significant effect on cooling characteristics, and the gas film cooling effectiveness obtained with curved surfaces produces enhanced results compared to those obtained with flat surfaces.

This review highlights the significance of factors such as turbulence in the mainstream, coolant Mach number, mainstream Mach number, blowing ratio, and geometric parameters—including injector geometry, injector orientation, and surface curvature—on film cooling effectiveness in aerospace applications. Available experimental and numerical investigations demonstrate that, compared to other geometric configurations, the tangential injection technique enhances higher film cooling effectiveness when used in liquid film cooling applications. Hence, this review recommends the tangential injection technique

for investigating hypersonic liquid film parameters, such as film cooling efficiency, film cooling coverage length, and cooling layer homogeneity, using the SPH method.

2.3. Current Trends and Developments in Film Cooling

The practicality of using gaseous film cooling to reduce heat flux in conventional and cutting-edge rocket propulsion systems is outlined in this section. The progress from the 1960s to the 2020s, covering experimental, numerical, theoretical, and analytical methods, is reviewed and summarized below.

In 1951, a pioneering experimental investigation was conducted at JPL to study how thermal flux distribution influences film cooling performance in rocket thrust chambers [9]. The propellant used in this study was JP-3 jet fuel. Similarly, an experimental investigation on the internal film cooling performance of a rocket reaction chamber, conducted using a propellant combination of liquid ammonia and liquid oxygen, revealed that in the case of a rocket engine wall fabricated from a porous material, the coolant can be ejected through the wall to create a protective film on the inner surface [29]. It was observed that the limitation of this cooling technique lies in the difficulty of achieving a steady flow through large sections, particularly when liquid coolants are employed as a secondary flow. In 1954, an experimental study on film cooling led to the development of a method for evaluating evaporation flux and wall temperature [32]. This method was used in investigations involving non-reactive liquid coolants for film cooling a wall exposed to steady high-speed gas flow in a duct. The study also investigated the influence of mass injection on transport processes within the turbulent central region. In 1964, an investigation aimed at identifying factors affecting the effectiveness of gaseous film cooling, such as multiple injector configurations, was conducted experimentally [77]. The study focused on the cooling efficiency of a typical nozzle employed in aerospace propulsion systems. In 1969, a study combining experimental and analytical methods investigated the viability of film cooling in rocket engines with earth-storable, space-storable, and cryogenic propellant combinations [36]. The findings revealed that using these propellants in combination achieved effective film cooling. In 1976, a study on flow coupling phenomena was conducted to experimentally investigate the feasibility of film cooling for thermal surface management and surface thermal treatment [78]. In the case of thermal surface management examined at the inlet and outlet of the flow, it was observed that the temperature profiles exhibited inflated values compared to corresponding analogies. In 1977, a thermal-mass transfer correlation technique was employed to experimentally examine the influence of flow property gradients, such as high-density gradients, on film cooling efficiency [79]. This investigation evaluated different injection configurations, and the findings revealed that the density ratio critically affects film cooling efficiency when coolant was injected through holes. In 1988, an experimental investigation using a wind tunnel was conducted to examine thermal energy transfer and thermal film robustness in a De Laval nozzle [22]. It is worth noting that the experimental setup for this investigation was powered under equilibrium conditions.

From 1991 to 1998, experimental studies advanced the understanding of interactions between the mainstream gas and injected coolant, particularly the influence of coolants on cooling efficiency. An experimental investigation quantified the impact of low-intensity swirling in the free stream gas on film cooling efficiency [76]. Results demonstrated that flow recovery severely disrupted the cooling jets, leading to markedly reduced cooling efficiency. Another experimental investigation examined the effects of flow property gradients, such as velocity and density, induced by the interactions between the mainstream flow and the injected coolant on film cooling efficiency [80]. The findings revealed that these flow property gradients significantly affect cooling efficiency, particularly when the investigation is conducted in a 2D configuration. An experimental study investigated the effects of

coolant injection parameters on film cooling performance [70]. Coolants such as air and helium were injected under supersonic conditions. The results revealed an overall enhancement in cooling efficiency that corresponded to an increase in the coolant's Mach number. However, it was observed that helium produced significantly higher cooling effectiveness compared to the scenario where air was employed. In a study aimed at experimentally enhancing and evaluating the optimal thrust generated by small propulsion nozzles, such as baseline conical and bell nozzles, various percentages of fuel were used as film coolant at known combustion chamber pressures [45]. Another experimental investigation was conducted to analyze the role of supersonic shock waves on film cooling effectiveness [56]. The findings showed the presence of momentum transfer between the mainstream and secondary flow at the interface. Film cooling performance was experimentally evaluated using a rocket propulsion test rig [73]. The experiments investigated the influence of flow parameters, such as coolant Mach number and temperature, as well as heat flux coefficients, on thermal efficiency. In 2008, a study investigating film cooling effectiveness was conducted using a flat plate with slot-injected coolant, employing both experimental and computational methods in non-turbulent supersonic gas streams [68]. The findings suggested that as long as the non-turbulent boundary layer remains weakly perturbed, the film cooling performance is independent of the injection momentum normal to the wall. In 2014, an experimental study evaluated CFD software (LOCI-CHEM, a RANS-based numerical simulation tool) for film cooling effectiveness in rocket thrust nozzles [81]. The investigation focused on supersonic airflows where the primary and secondary flows were air, and streamwise coolant injection techniques were employed. In the case of thermal transfer rate measurements, the RANS-based CFD method was found to be reliable in evaluating the thermal transfer rate of the downstream film-cooled wall, and the technique significantly overestimated the thermal transfer rate on the upper adiabatic wall.

Subsequently, a review of the numerical methods studied over the years is presented below. From 1992 to 1994, numerical studies explored adiabatic thermal gradients, slot injection cooling techniques, and the impact of wake turbulence and boundary layers on cooling performance. The influence of protective effects on the cooling effectiveness and adiabatic thermal gradients in a supersonic nozzle [71] was numerically investigated in 1992. The study employed a backward-configured slot for coolant injection, and the numerical findings confirmed the feasibility of film cooling in supersonic regimes. In 1993, a numerical study investigated various problems encountered in slot injection cooling techniques employed in propulsion system design, and the simulated findings were evaluated using experimental data [48]. A finite element fluid dynamics solver was employed to investigate internal fluid dynamics problems, with a focus on film cooling efficiency. The numerical findings were subsequently validated using experimental data. The study proved the viability of employing numerical techniques in investigations involving CD nozzles, provided the scope of constraints is clearly defined. In 1994, a numerical study examined the effect of wake turbulence on film cooling efficiency using 2D dynamic and time-independent simulations [82]. The numerical findings showed that dynamic simulations are necessary to capture the fluid flow distribution. In 1998, a rocket propulsion system employed for an experimental study investigated the cooling effectiveness using a boundary-layer computational tool [73]. The study utilized compressibility factor modification techniques and different turbulence simulation methods for model validation. During the 2000s, numerical studies contributed to a deeper understanding of key parameters, including flow instabilities, thermal radiation rates, flow entrainment rates, coolant injection techniques, shock-induced separation, side-load effects, shock wave interactions, thermal transfer rates, and boundary layer flow profiles [9,49,83,84]. In 2002, a sub-component LE-7A propulsion nozzle, comprising different designs, was utilized to investigate the role

of flow instabilities induced in the unsteady flow by nozzle surface configurations [83]. Numerical investigations were necessary to examine the underlying factors of the side loads detected during an experimental study involving this model.

In 2003, a Navier–Stokes flow solver was used to analyze the thermal radiation rate in a thrust nozzle [49]. The findings revealed that, when approximating wall temperature, the thermal radiation rate must be considered for determining film cooling effectiveness in high-capacity thrust nozzles due to significant temperature variations. Another study involving a large eddy simulation (LES) method examined film cooling performance on an idealized turbine blade surface [84]. The investigation focused on the influence of flow entrainment rates, including the mixing mechanisms occurring in the wake flow zone, based on geometric factors such as surface area, mean curvature, and other factors. Researchers established a database in 2006 to validate various computational fluid dynamics (CFD) techniques for film cooling studies [85]. The database focused on two geometrical configurations: a flat plate and a semi-cylindrical leading edge, and detailed descriptions of the CFD results obtained for these models. In another study, the direct numerical simulation (DNS) method was utilized to investigate the cooling performance involving a turbulent flow model [86]. It was observed that the variables in the k and ε equations, which incorporate pressure, significantly affected the coolant jet. In 2008, a pioneering study numerically determining film cooling performance on a 2D wall subjected to slot-injected coolant and eddy mixing was conducted using the Large Eddy Simulation (LES) model [87]. Although the numerical findings revealed a restrictive assumption in the context of constant-density validation, scale-dependent sub-grid configurations provided deeper insights into turbulent diffusion processes. Steady-state and transient computations were conducted to investigate the effect of shock-induced separation movement on film cooling effectiveness in a nozzle segment, and the numerical findings were evaluated experimentally [88]. The experimental findings revealed the presence of oscillatory motion in the energy dissipation system due to the film stagnation pressure. In 2009, a numerical study of side loads was conducted on a film-cooled nozzle segment [50]. The investigation revealed turbine combustion products as the primary source of side-loading. Similarly, a numerical performance analysis was conducted for an enhanced nozzle segment by film-cooling the nozzle wall [47]. The results revealed that the resultant film cooling effect introduced an increasing inflection point and length, thereby increasing the coolant discharge rate. Similarly, a numerical method was employed to investigate the influence of shock waves on film cooling performance, comparing the results with scenarios where no shock waves were present under similar supersonic flow conditions [75]. It was determined that shock waves in supersonic flow conditions demonstrated a considerable influence on low-density gasses compared to high-density gasses. In 2010, a numerical investigation aimed at forecasting the thermal behavior of rocket propulsion systems employed a RANS-based solver to evaluate the thermal transfer rate near the injector surface [89]. It was observed that this particular technique captured the consistent pattern identified in the experimental heat transfer analysis. Similarly, a RANS-based computational technique was employed to predict the physical processes influencing film cooling effectiveness under high-speed turbulent flow conditions [90]. The findings revealed that the accuracy of this computational technique depends on the fidelity of the target outcomes. The method proved insufficient for cases such as mixing near the injector region, including scenarios where high-precision film decay rates are a priority. In 2011, a CFD RANS-based computational method was introduced to evaluate the effects of flow parameters, such as combustion pressure, cooling slot configuration, blowing ratio, and temperature and velocity ratios, on film cooling effectiveness involving the injection of cryogenic propellant [91]. The study affirmed the reliability of the solver employed and the viability

of the selected turbulence model. In 2012, a numerical technique based on a reduced-dimensional integral formulation was used to investigate the evolved flow pattern of a wall-bounded film jet within combustion chambers [58]. This method was more reliable in estimating the transformation of the injected coolant, surface temperature profiles, and film cooling efficiency.

Additionally, investigations employing theoretical methods are reviewed and presented below. A theoretical investigation on the effect of compression and non-isothermal behavior conducted in 1969 revealed that these parameters commonly applied in practical aerospace applications have no significant impact on film cooling effectiveness [92]. However, over time, researchers have used various experimental and numerical techniques to validate this finding, and deeper insights into this phenomenon have led to the development of advanced thermal protection systems. The prior investigation led to a similar study in 1970, focusing on the rapidly developing turbulent viscous layer on a thermally insulated flat plate extending downstream of a gaseous coolant injection point [93].

Studies addressing other factors are reviewed and presented below. In 1987, a review was conducted on the role of transient side loads in engine failures observed in the Space Shuttle Main Engine (SSME) [51]. Although a large portion of the recorded failures occurred due to the poorly defined nature of transient loads, access to formal reports was restricted. The review, therefore, failed to meet the required standards [9]. In another investigation conducted in 2002, a large side load was identified as one of the contributing factors to engine failure in the LE-7A engine [52]. The findings revealed that this phenomenon occurred during power-up and power-down sequences. The induced side loads included the Do you have any updates about how you eventually resolved the issue? Thank you!(FSS) and the Restricted Shock Separation (RSS). The large side loads arose from asymmetric pressure distributions during transient phases, causing uneven forces on the nozzle structure. These forces increased the risk of nozzle deformation or fatigue failure, potentially leading to catastrophic engine failure. The key mitigation strategies include nozzle design modifications, optimization of the power-up and power-down sequences, active combustion control, and material upgrades. After modification, the LE-7A demonstrated reliable operation, with no reported side-load-related failures. In 2012, a study investigated the techniques for creating and applying appropriate models for film cooling applications in rocket reaction chambers [54]. The study employed quantitative and semi-theoretical correlations, demonstrating that the pattern observed in the NASA SP-8124 liquid film model lacked a satisfactory explanation. Additionally, the model adopted by Stechman was found to be incompatible with the given application. In 2020, artificial intelligence (AI) was developed to evaluate film cooling efficiency in an air-cooled turbine vane [94]. The model analyzed the temperature profile on the vane surface when the film coolant holes were altered. The developed AI model showed the ability to estimate film cooling performance on the exterior surface of an air-cooled turbine vane. In 2024, a study investigated the influence of nanofluid coolant on film cooling performance [95]. The parameters examined included particle size characteristics, particle structure, nanoparticle concentration, and temperature profile.

The trends and advancements in film cooling technology utilized in aerospace applications have identified various unsolved problems and research gaps associated with this technology. These research problems are presented as follows:

(1) It is nearly impractical to accurately analyze the rapid transition from one operating mode to another in dual-bell nozzles because the intricate flow field induced in these components has not been fully understood. Therefore, investigations involving numerical methods are necessary to fully comprehend the complex flow interactions between the injected coolant and the primary flow field;

(2) The factors caused by side loads in film cooling investigations during transient operations in aerospace applications have not been fully understood. Therefore, research on advanced numerical techniques is necessary to address this issue;
(3) In the numerical investigations of the cooling performance in a rocket propulsion system utilizing the boundary-layer computational technique, it was observed that an accurate approximation of the turbulent mixing region is essential for investigating the complex flows in supersonic regimes. Therefore, the SPH method is recommended for accurately modeling the turbulent mixing region due to its numerous advantages in handling intricate flow phenomena;
(4) The difficulty in numerically validating the experimental study, involving thermal transfer rate measurements using the RANS-based CFD method in rocket thrust nozzles, revealed the need for further studies, particularly focusing on the mechanisms of flow dynamics in the upper adiabatic wall of the nozzles. Thermal transfer rate measurements using advanced Schlieren technology, focusing on turbulent structures along the upper adiabatic surfaces, will provide insights into understanding this flow phenomenon;
(5) Further studies on nanoparticle surface alterations, including surfactant application, are highly recommended for future investigations. These studies will explore the factors critical to nanofluid dispersion stability for real-world applications.

The methods employed by researchers and designers over the years to advance film cooling technology for aerospace vehicles are presented below. An experimental investigation was conducted on multi-jet film cooling utilizing a standard nozzle and coolant gasses [9]. The findings showed that multi-jet gas film cooling is among the most efficient methods for cooling with high-energy propellants. No erosion was observed at the nozzle throat using methane and hydrogen, while carbon monoxide resulted in minor erosion. In their experimental research on film cooling, Back and Cuffel [78] examined the influence of wall cooling on the flow dynamics, shockwave patterns, and degree of flow detachment in the interaction between shock waves and turbulent boundary layers. They discovered that the detachment area was smaller when the film cooling technique was employed, in contrast to the scenario without cooling. Additionally, they observed that the detachment and reflected shock waves merged near the boundary layer edge, continuing into the free stream as a single wave. Gau et al. [76] conducted experiments in a film-cooled circular pipe of a specified sudden expansion ratio. The results indicated that the rotational flow in the mainstream greatly influenced film cooling performance.

Stoll and Straub [22] employed a parabolic numerical solver for boundary layer analysis using the k-ε turbulence model to study film cooling and thermal transfer in nozzles featuring smooth, rounded throat geometries. They observed that the throat region did not affect the film's stability. They also emphasized that the mass flow rate ratio of coolant to mainstream led to a notable reduction in heat transfer in both subsonic and supersonic regimes. Arrington et al. [45] evaluated the thermal protection performance of the cooling film in a baseline conical nozzle compared to a bell nozzle, using gaseous H_2 and O_2 as propellant fuels. The study demonstrated the successful application of film cooling in both nozzle designs. Martelli et al. [47] performed a numerical analysis on the impact of gaseous film injection in a dual-bell nozzle. They discovered that the inflection point led to an expansion, which lowered the surface recovery temperature and minimized mixing effects. As a result, improved wall protection was achieved due to the expansion fan created by the inflection point. Matesanz et al. [48] conducted numerical simulations of slot injection cooling in convergent-divergent (CD) nozzles. Their findings predicted adiabatic wall temperatures and film cooling effectiveness, which aligned closely with the corresponding experimental data. These results demonstrated the viability of

gas-based film cooling techniques for reducing thermal loads in both conventional and next-generation rocket nozzles. However, the effect of film cooling on nozzle performance, particularly concerning flow separation and related side loads, may impose limitations on this technique, as evidenced by the documented nozzle wall failure during preliminary sea-level testing of the LE-7A engine [52]. Takahashi et al. [83] numerically examined time-dependent flow in a subscale LE-7A nozzle using a thermal barrier film.

In propulsion systems, the impact of thermal barrier films on separation dynamics during transient startup and shutdown phases can vary substantially, depending on the interaction between the free-stream flow and the secondary coolant flow dynamics [9]. Reijasse and Boccaletto [88] conducted experimental and numerical investigations into the impact of film cooling on nozzle flow separation. Their research uncovered a dynamic occurrence in the shock-induced separation zone during high-speed coolant injection. Wang and Guidos [50] conducted a time-dependent numerical analysis of transverse loads in a film-cooled nozzle expansion section. Their simulations focused on the engine ignition and shutdown phases. The results indicated that peak side loads were generated because of Mach disk flow and the resulting shift in the separation line. While the steady-state operation is predictable, the impact of film cooling under transient conditions remains unclear. A comprehensive analysis of the side loads produced during transient operation is essential, as these loads are crucial for assessing the viability of implementing film cooling in rocket nozzles during transient phases [9]. Similarly, the theoretical assessment of how velocity and temperature compressibility influence the thermal protection performance of the cooling film revealed that density variation impacts are of critical significance but can be considered negligible in real-world combustion chambers, as reported by Volchkov et al. [92]. Likewise, Repukhov [93] examined the impact of compressibility using a turbulence model based on boundary layer equations and obtained comparable results. Hansmann et al. [80] and Pedersen et al. [79] demonstrated that compressibility can notably influence thermal protection performance, with their findings showing increased effectiveness at higher coolant-to-mainstream ratios. O'Connor and Haji Sheikh [71], along with Kuo et al. [72], studied the influence of primary flow velocity in a film cooling duct. Kuo et al.'s experimental findings indicated that the film cooling technique was effective in subsonic applications but not under supersonic conditions. Under supersonic conditions, they noted that the injected flow tended to bend upstream, counteracting the upstream flow. Juhany et al. [70] observed an increase in cooling effectiveness as the heat capacity of the gas rose, based on experimental comparisons between helium and air. The study also demonstrated that cooling effectiveness improved with higher coolant Mach numbers. An experimental investigation of supersonic film injection conducted by Aupoix et al. [73] explored two different injection geometries and pressure ratios. Their results showed good agreement with those of Kuo et al.'s experimental findings. Kanda et al. [56] examined the behavior of film cooling under external shock conditions in a supersonic wind tunnel. They observed a reduction in film cooling effectiveness at higher pressure ratios, which was directly attributed to a decrease in the flow Mach number at a specific location. Peng and Jiang [75] performed a numerical study that showed a reduction in adiabatic effectiveness caused by the impinging oblique shock wave, thereby validating earlier results. An investigation on film cooling in the laminar supersonic regime found that cooling effectiveness improved with higher coolant mass flow, while slot height had little influence [9]. Heufer and Olivier [68] conducted experimental and numerical studies on film cooling using the slot injection method on a flat plate in non-turbulent supersonic streams.

Both experimental and numerical findings indicated that the blowing parameters did not affect cooling effectiveness. The cooling effect was dependent on the mainstream gas flow properties, and the film cooling technique was reported to be very effective

in laminar flow conditions. Jansson et al. [82] employed the turbulence kinetic energy-dissipation model and an algebraic Reynolds stress approximation to conduct steady-state and transient computational studies. They observed that the time-averaged flow and temperature distributions were accurately predicted for the steady-state case, whereas the temperature predictions varied considerably in the transient scenario. The Spalart–Allmaras (SA) turbulence approximation and the RANS-based solver have been utilized by Cruz, Cruz, and Marshall, Betti and Martelli [87,89,96], among others. The SA model with a constant turbulent Prandtl number produced mean temperature and velocity profiles that closely match the experimental values [9]. Voegele et al. [90] utilized the LOCI-CHEM RANS solver to simulate a film-cooled wall, incorporating turbulence flow approximations such as the two-equation shear stress transport (SST) model and the one-equation Spalart–Allmaras (SA) model. They reported that the SA model accurately predicted the initial decay region, while the SST model aligned closely with experimental values for the wall wake. However, neither model successfully predicted the film decay rate in the outer flow domain. Na et al. [85] employed a volume-integrated numerical method with a second-order upwind differencing technique and an eddy viscosity and diffusivity formulation method to investigate film cooling. They noted that the SST turbulence model provided more accurate predictions for a flat plate configuration, and the turbulence kinetic energy-dissipation model was better suited for a semi-cylindrical leading edge with a flat tail-end design. Badinand and Fransson [49] conducted a computational analysis to assess the significance of radiative heat transfer in a film-cooled nozzle. Their findings revealed that in the nozzle with embedded shocks, the emission from the shock wave disk is as strong as that emitted from the reaction zone. Additionally, it was observed that the thermal radiation flux is also significant. The Reynolds-averaged Navier–Stokes (RANS) code and the Cooling Power Simulator (CPS) developed by ONERA, Bertin Technologies, and CNES were employed for numerical analysis [88]. In this study, the effect of film cooling on boundary layer separation was investigated experimentally and numerically. The numerical results aligned with experimental observations and revealed the movement of the separation shock foot during the injection of the film coolant. Betti et al. [91] developed a 3D RANS solver adapted to model multi-component mixtures of thermally perfect gasses. This model was used to analyze coolant film injection in cryogenic propulsion systems. Di Matteo et al. [58] utilized EcosimPro, an object-based computational platform, to create a simplified two-dimensional integral approach for examining a film wall jet in reaction chambers. They validated this model using test results from a DLR H_2/O_2 reaction chamber. Maqbool et al. [81] conducted experiments to validate the coolant layer efficiency predictions using LOCI-CHEM, NASA's thermal flow modeling tool. These models can assist in the preliminary design of cryogenic propulsion systems, evaluate film cooling effectiveness, and forecast performance losses. Film cooling investigations were conducted in a converging-diverging nozzle using a finite element Navier–Stokes CFD code with simulation methods such as Large Eddy Simulations (LESs) and Direct Numerical Simulations (DNSs) [48]. While the predictions were quite accurate, they were restricted to a limited number of comparisons. Similarly, Tyagi and Acharya, as well as Muldoon and Acharya [84,86], employed LES and DNS methods to predict key flow parameters and heat transfer characteristics. Transient side loads on nozzles have been observed in various rocket engines during their early development, posing a risk of structural damage [51,52,97]. When using TEG to cool the nozzle extension, additional side load dynamics are introduced due to the impact of the coolant layer dynamics on the normal shock wave structures [9]. It is noteworthy that initial hypotheses regarding the prediction of film cooling effectiveness are derived from the assumptions put forth by Librizzi and Cresci [77]. They hypothesized that the thermal management fluid is completely mixed with the mainstream within the

near-wall layer, resulting in a uniform temperature, with any discontinuity occurring solely at the outer edge. They also observed that the growth of the boundary layer follows typical patterns and is not affected by the injection.

It has been observed that turbulence effects between the thermal protection layer and the mainstream lead to entrainment, making it relevant to gaseous film cooling [9]. Kirchberger et al. [54] summarized the quantitative and semi-theoretical correlations for assessing film cooling efficiency in the open literature, including their application in a GOX/kerosene reaction chamber. Their findings indicate that many facets of cooling through gaseous boundary layer formation are now being uncovered. An experimental engine operated at a thrust of 4448 N was utilized for film cooling investigations [9]. Various film coolants, including water, aniline-alcohol fuel, 60-octane gasoline, methyl alcohol, and anhydrous ammonia, were used to study the effects of film cooling efficiency. The experimental results indicated a reduction in heat flux at the engine walls. Similarly, coolants such as water, ethyl alcohol, and liquid ammonia were employed to examine film cooling in a rocket engine using liquid ammonia and liquid oxygen as propellants [29]. These experiments were instrumental in estimating the film coolant requirements for cylindrical combustion chambers in rockets with similar propellant injectors. Knuth [32] established the conditions for the stability of a thin liquid film flowing under the influence of high-velocity gas streams exhibiting turbulence. Stechman et al. [36] examined the applicability of thermal protection via film cooling in rocket propulsion systems that utilize earth-storable, space-storable, and cryogenic propellant combinations. They concluded that these propellants are highly suitable for this application due to their ability to maintain a liquid phase, enhancing efficiency. These propellants also enhance film cooling effectiveness due to their significant latent heat of vaporization. The research on liquid film dynamics in a baseline nozzle throat found that the most effective cooling occurred when the liquid was injected directly into a linearly tapering segment and at the throat [9]. Research has shown the limitations of regenerative coolants used in rocket propulsion systems. Carbon film formation was the main limiting factor for hydrocarbon fuels, as observed by Cook and Quentmeyer [53]. Volkmann et al. [46] investigated how film cooling impacts the reduction in heat transfer intensity at the throat of a spacecraft propulsion system. Their experiments were conducted using a subscale LOX/RP-1 high-pressure reaction chamber. They noted that film cooling resulted in a peak reduction in heat flux. Kirchberger et al. [55] conducted film cooling experiments using a small-scale heat-absorbing test model with GOX, nitrogen, and kerosene as coolants. Their findings indicated that kerosene, as a coolant, was significantly more effective for film cooling than nitrogen. Gater et al. [57] conducted experimental and computational studies on film cooling techniques using a flat film coolant to assess the liquid quantity attached to the wall using a sharp-edged retention slot. They concluded that although liquid-film cooling in rocket propulsion chambers has proven viable, the underlying phenomena remain poorly understood. Therefore, further investigation was suggested to understand flow mechanisms such as film disruption in the liquid film, droplet entrainment, and film surface irregularities, which enable improved cooling performance for this application. The experimental and computational findings have revealed that the effectiveness of film cooling techniques varies with the blowing ratio, a phenomenon attributed to changes in the strength of vortex structures downstream of the coolant injection point and the jet exit momentum [44]. The study revealed a specific ideal blowing parameter for each geometric configuration.

Research on film cooling has led to significant advancements, including developments in cooling techniques, insights into heat transfer enhancement, advancements in numerical simulations, insights into the effects of surface roughness, and the introduction of innovative coolant materials. Machine learning techniques have been utilized to forecast the

cooling effectiveness across a complete 3D guide vane using the geometric parameters of the film cooling structure, offering both speed and accuracy [94]. This method enhances the initial stages of turbine design by employing AI technology. The thermal energy transfer coefficient and flow behavior characteristics of two types of oxide nanofluids, specifically CuO and Al_2O_3, were systematically explored by Sun et al. [95]. Their investigations were conducted within a standard commercial data center environment that uses liquid thermal management fluids. The findings confirm that incorporating nanoparticles into the film cooling process can greatly enhance the thermal transfer efficiency of hydrocarbon-based coolants. França et al. [98] investigated the thermal conductivity of ionic liquids and nanofluids to evaluate their potential as heat transfer fluids. Their findings revealed that incorporating nanomaterials into cyano-based ionic liquids significantly improved thermal conductivity. The advancement of sophisticated computational models, such as CFD simulations, has enabled deeper insights into the intricate dynamics of film cooling flows and associated heat transfer processes [99]. Four coolant channel configurations were simulated to determine the impact of internal flow structures on external film cooling performance. The experimental data for the smooth and the angled ribbed channel closely align with the numerical results. Large eddy simulations (LESs) were conducted to examine how surface roughness, applied to the internal linings of a cooling hole, affects thermal management efficiency and flow structures [100]. The findings showed that increasing the surface roughness of the cooling hole thickens the fluid interface layers inside the hole, particularly at higher blowing ratios. Goldstein et al. [101] conducted an experimental study on film-based thermal management over a contoured surface using a mass transfer approach. The study employed two different injection geometries, one featuring a single row of holes and the other featuring two rows. Their findings revealed that the contoured surface reduced the spanwise-averaged effectiveness for both geometries at low coolant injection rates. Moon et al. [102] evaluated the performance of a gas turbine combined cycle by employing coolant inter-cooling (CIC) and coolant pre-cooling (CPC) techniques. To enhance efficiency, the system incorporated heat recovery from spent coolant to reclaim the heat lost during the cooling process. The key findings are as follows: The CIC method provided a larger power increase for the gas turbine than the CPC method, as it demands less energy for coolant compression. Overall, the combined cycle power plant (CCPP) showed superior performance with the CIC method compared to the CPC approach.

These advancements in film cooling research highlight critical areas of progress, contributing to a deeper understanding and development of cooling technologies in various applications, including gas turbines and heat exchangers. The review demonstrates the practicality of employing liquid film cooling techniques in spacecraft propulsion systems using various propellant-oxidizer combinations and the viability of gaseous film cooling techniques in experimental and numerical applications, particularly for nozzle cooling [9]. It also highlights that the various flow parameters have a significant influence on film cooling performance.

2.4. Computational Fluid Dynamics (CFDs) for Film Cooling

Modeling film cooling poses challenges like turbulence, heat transfer, and boundary layer interactions. To address these challenges, CFD methods such as RANS, LESs, DNSs, and hybrid approaches are used in film cooling applications [9]. Another CFD method used in film cooling investigations is the Smooth Particle Hydrodynamics (SPH) method, which offers several advantages over other CFD methods.

Progress on the SPH method is reviewed and summarized below. A review of the principles and applications of the SPH method presented detailed deliberations on this CFD technique, from its origin to a comparison with particle motion, as well as the var-

ious applications where the SPH method has been successfully employed, including its advantages and disadvantages [103]. Another overview of the SPH method discussed the key aspects of the SPH technique, including its latest advancements, the necessity for meshless particle techniques, the significance of smoothing kernel functions, and a broad strategy to develop smoothing kernel functions, among others [104]. An investigation into the data organization employed in DualSPHysics has transformed its application into a concurrent framework [105]. The study outlined a methodology for significantly increasing the performance of the SPH method. Similarly, a review highlighted the current progress in the SPH method, focusing on its application in multiphase flow [106].

This review presents CFD methods for analyzing film cooling effectiveness in combustion chambers, turbine blades, and hypersonic vehicles. The various CFD techniques for advancing aerospace film cooling technology are reviewed and summarized below. Monaghan [103] provided an in-depth review of the key benefits of SPH, highlighting its mesh-free nature, which is particularly useful for simulating free-surface flows and large deformations without re-meshing. SPH's ability to manage complex geometries and dynamic interfaces makes it well-suited for simulations involving free surfaces and significant deformations [104]. It accurately simulates interfacial and free-surface flows, underscoring its effectiveness in capturing complex fluid behaviors [107]. Initial comparisons of the SPH method mainly concentrated on its use in astrophysical applications [108]. However, a foundation has been established for understanding its benefits over grid-based methods in managing complex, deformable systems. Long et al. [105] developed a high-performance domain splitting method and redesigned the data structure of the DualSPH numerical platform to work within a concurrent processing framework. Their comprehensive testing shows that, with 3 to 120 million particles, the system maintains more than 90% computational efficiency, highlighting the superior performance of the parallel strategy.

Wang et al. [106] examined recent progress focused on improving the accuracy and stability of SPH, addressing some of the common challenges historically linked to the method, such as enhancing accuracy by compensating for boundary deficiencies. Similarly, Skillen et al. [109] introduced techniques to improve the effectiveness of SPH in simulating incompressible flows, body-water impacts, and wave-body interactions. They found that these techniques can lead to temporal pressure noise in certain applications, although spatial noise is largely removed. The findings are precise and nearly free of noise in both space and time, indicating successful convergence. It has been proven that the choice of these CFD methods typically depends on the complexity of the problem and the desired level of accuracy in the results. Xia et al. [110] comprehensively reviewed the numerical techniques employed in effusion cooling studies. Their investigation revealed that empirical models often rely on fundamental principles and conservation laws to solve the RANS equations, incorporating higher-fidelity techniques such as LES and hybrid RANS-LES approaches, including Detached-Eddy Simulation (DES). The study primarily concentrated on applications mainly associated with the internal flows of gas turbine engines. Early studies primarily used RANS-based computations combined with the k-ε turbulence model. The predominant computational methods currently rely on RANS, employing eddy-diffusivity or Reynolds stress models. In contrast, LES and DNS techniques directly incorporate the turbulent scales of the flow, potentially making them more suitable for film cooling applications when compared to the fully modeled turbulence approach of RANS [9]. Software tools commonly used for CFD in film cooling include ANSYS Fluent (versions 18.0–19.2 and 2020 R2–2023 R2), OpenFOAM (v10 and v11), STAR-CCM+ (2021.3 and 2022.1–2023.2), and COMSOL (6.0 and later) Multiphysics, which are widely employed for simulating fluid flow, heat transfer, and turbulence in complex geometries. Future

directions include machine learning for turbulence modeling, high-fidelity LES and DNS for accuracy, and the optimization of CFD-driven cooling holes.

3. Film Cooling for Combustion Chamber

The combustion chamber must balance efficiency, durability, thermal management, and stability under operational and environmental constraints. Studies have focused on advancing parameters such as blowing ratios [68], coolant Mach number [70], external shockwaves [8], freestream turbulence [40], injector orientations [43], and others to achieve these key requirements.

Section 3.1 reviews techniques for addressing film cooling challenges in aerospace combustion chambers. Similarly, Section 3.2 addresses key innovations in rocket propulsion chamber film cooling.

3.1. Studies on Combustion Chamber Film Cooling

The intense heat produced during the combustion process creates significant challenges in maintaining the durability and lifespan of the combustion chamber walls. Hence, effective cooling mechanisms and the film cooling technique can help reduce thermal damage by shielding the combustion chamber from excessive heat. The process by which aerospace vehicle thrust chambers are film-cooled begins with pumping fuel or oxidizer through channels that surround the combustion chamber walls in regenerative cooling [111]. As it flows through these channels, the fuel or oxidizer absorbs heat, cooling the chamber walls and enhancing the system's efficiency. The heated fuel or oxidizer exits the cooling channels close to the injector face. The film cooling technique used in combustion chambers offers advantages such as improved component durability and performance, making it indispensable in high-temperature applications like gas turbines.

The recent progress in film cooling technology used in combustion chambers is summarized below. In 1990, NASA presented an outline of the various cutting-edge launch system projects necessary for developing the cooling technology required to support enhancement-driven approaches [112]. Film cooling effectiveness in a liquid fuel combustion system is essential. Hence, in 2005, various cooling methods, including regenerative, film, dump, transpiration, and radiation cooling, were employed in such systems [113]. In 2007, a regenerative cooling technique involving a three-step investigation approach using subcritical and supercritical coolant streams was employed to develop an appropriate analytical and simulation tool capable of describing the heat transfer characteristics inside the cooling channels of a liquid propellant rocket system [114]. It is worth noting that the simulation model was validated using data sourced from published literature. In 2014, studies on regenerative cooling techniques used in rocket propulsion combustion systems revealed that integrating High-Aspect-Ratio Cooling Channels (HARCCs) into the cooling system design increased the engine's lifespan by reducing the thrust chamber wall temperature [115,116]. Regenerative cooling is considered one of the most promising thermal protection methods for hypersonic vehicles; however, a 2017 investigation revealed that uneven flow distribution within the cooling channels can lead to combustion chamber overheating [117]. In 2021, time-independent and time-dependent numerical simulations were conducted using the ANSYS Fluent CFD code to analyze the time-varying characteristics of rocket combustion during the start-up sequence, with fuel injection performed at subcritical pressure and temperature conditions [118]. In 2024, a simplified 2D configuration of a cooled combustion chamber injector was used to numerically investigate the effects of the mass ratio and the development of the coolant film [6]. The study also examined the influence of heat transfer characteristics on the performance of film cooling in the combustion chamber. The

numerical findings, validated through experiments, confirmed the viability of this CFD technique for the application.

Over the years, researchers have employed various cooling methods to address the numerous unsolved problems associated with combustion chamber cooling technology. These techniques are reviewed and summarized below. NASA conducted a study to examine various enhancement-driven design approaches, such as cost-reduction strategies for chamber manufacturing and challenges in hydrocarbon fuel compatibility with combustion chamber liner materials [112]. The findings revealed that modern engines, possessing much greater core strength than first-generation gas turbines, can reach temperatures above 1700 °C. Carlile and Quentmeyer [119] conducted an experimental study to examine the impact of elongated cooling channels on the cooling performance of high-thrust rocket chambers. Another key area of the experimental investigation included life cycle enhancement and a reduction in coolant pressure differential. The findings suggested that the wall surface thermal state can be lowered by significantly enhancing the contact area on the coolant side, in contrast to the hot-gas side. The dynamic response of a rocket combustion chamber was examined during the ignition phase using the coaxial injector shear interaction test case [118]. The numerical findings, when compared to experimental data, revealed significant differences. These errors were attributed to inaccurate modeling of the state equation, as well as incomplete data on initial and boundary parameters. The temperature and pressure inside the combustion chamber are crucial factors in selecting the appropriate cooling technique. Therefore, another investigation employed numerical methods to study the flow characteristics in a liquid fuel combustion chamber using high-pressure, high-temperature hydrogen coolant [113]. The numerical results were compared to experimental data to analyze fixed and dynamic properties. The results revealed that the velocity field experienced a more significant influence on the temperature field due to considerable density variations in the high-pressure, high-temperature hydrogen coolant. Jing et al. [117] conducted a computational study on the fluid dynamics and heat transfer characteristics of the coolants employed in combustion chambers. One of their findings revealed a considerable distribution of the mass discharge rate, corresponding to a significant pressure drop caused by the presence of a curved inlet manifold. Studies have proven that heat and mass transfer between the coolant film and the combustion gasses cause the coolant film's effective thickness to decrease in regions with lower temperatures, thereby directing extra coolant to one or more downstream chambers [111].

Different investigations have revealed that the cooling system's performance is crucial in enhancing the cooling technology for liquid fuel combustion chamber applications [6]. Therefore, a regenerative cooling approach was used to investigate the impact of the highest aspect ratio on rectangular cross-sections. Initially, cooling channels with a circular cross-section were used. Nevertheless, rectangular cross-sections were subsequently chosen because of their better performance in lowering wall temperatures [114]. In another regenerative cooling investigation conducted by Park [115], a turbulence simulation method was utilized to examine the turbulent fluid dynamics and heat transfer characteristics in a rocket propulsion system. It should be noted that the turbulence prediction models examined were subjected to varying shapes and proportions of the cooling passage under steady or fluctuating heat transfer rates at constant thermal mass flux. Meanwhile, at an increased aspect ratio, cooling performance decreased significantly. Subsequently, the benefits of employing a fluctuating channel aspect ratio in rocket propulsion combustion systems were examined numerically by Pizzarelli et al. [116] under defined limitations. The results revealed that the peak cooling performance was achieved at an aspect ratio exceeding the unit aspect ratio. The review has determined that the effective design of cooling channels employed in aerospace propulsion combustion systems depends on two main factors: the

temperature of the hot gas sidewall and the pressure drop across the channels, with the pressure drop ideally being minimized [120].

Another crucial factor enhancing the cooling performance of liquid-fuel combustion chambers is porosity. Therefore, an approximate theory of porous film cooling techniques initially developed for non-reactive coolants was extended to investigate the scenario in which one of the propellants was used as a reactive coolant [38]. The findings indicated that the temperature of the hot gas sidewall must remain within acceptable bounds. The review found that studies have explored combustion gas characteristics, gas composition, and enhancements in chamber shape as they relate to the liquid film cooling technique [9]. The effectiveness of film cooling using holes drilled into a trench has been experimentally examined and compared with the conventional cooling method without a trench [121]. The findings revealed that enhanced injection parameters resulted in more effective cooling compared to scenarios where other coolant injection techniques were used. Cruz and Marshall [96] investigated the gas calculations for a film-cooled wall surface. Their experiments included measuring wall temperature and adiabatic effectiveness and calculating turbulent flow across a range of precisely controlled inlet conditions. Mean flow temperature measurements in the far field exhibited a consistent pattern, while the temperature fluctuations suggested well-established turbulence in the wall layer. The liquid film cooling of a liquid rocket combustion chamber was conducted using the control volume method and the energy equilibrium principle [122], with experimental research on film cooling under various operating conditions with different film fluids. It was observed that the influence of parameters such as radiative heat transfer and coolant mixing was considerably higher. Optimal combustion performance was observed in experimental and numerical investigations involving methane as the fuel and oxygen as the oxidizer in a subscale combustion chamber [123]. The results indicated that the heat flux rate was regulated through the application of film cooling along the chamber walls; however, the role of the coolant film and inlet temperature on the cooling performance was negligible. Similarly, a performance analysis was conducted on a scramjet engine using hydrocarbon fuel as a coolant, employing both regenerative and film cooling techniques [124]. The findings revealed that the flow path within the cooling channel significantly influenced the performance of Radial Cooling (RAC) and Film Cooling (FC). A steady-state combustion simulation was validated by comparing the numerical results with experimental data [125]. The transient combustion simulation reveals that when the combustion frequency aligns closely with the chamber's modal frequency, the amplitude increases substantially, resulting in a high-intensity acoustic environment. Consequently, this leads to fuel savings. Over the years, investigations into film-cooling effectiveness of liquid propellant rocket engine combustion chambers have consistently demonstrated their capacity to provide high performance, dependability, and operational versatility throughout the modern era of rocketry [6].

3.2. Innovations in Combustion Chamber Film Cooling

The research on combustion chamber film cooling technology has led to the development of advanced thermal protection systems (TPS), capable of providing enhanced control over key variables such as blowing ratios, coolant Mach number, external shockwaves, freestream turbulence, and injector orientations [8]. By understanding and optimizing these variables, engineers can enhance film cooling performance, strengthen thermal protection, improve engine efficiency, and reduce fuel consumption. Each variable plays a role in overall cooling efficiency, and technological advancements optimize these parameters to improve performance in high-temperature, high-pressure conditions.

Studies have demonstrated that film cooling using low molecular weight gasses leads to a smaller reduction in specific impulse and could offer greater benefits [9]. Arnold

et al. [21] and Spalding [41] numerically investigated the influence of the temperature profile on film cooling performance in a nozzle section of a subscale thrust chamber, examining different Mach regimes. For film cooling effectiveness, the numerical results obtained for the surface of the CD nozzle section were compared with the experimental results reported by Shine et al. [44], as illustrated in Figure 5. Shine et al.'s experimental model yields significantly higher values than the numerical models. In contrast, the predictions from the other models are almost identical in the regions near the injection point. Xiang et al. [126] adopted an orthogonal methodology and the Kriging model to optimize the film cooling system through numerical simulation. The study focused on film cooling enhancement by varying critical parameters, including diameters, longitudinal and transverse angles, and hole configuration. Liquid oxygen (LOX) flows through rear-mounted manifold-fed posts, while H_2 injects via lateral orifices [69]. Tables 9 and 10 summarize the research on gaseous film cooling techniques for rocket propulsion chambers, focusing on experimental and numerical approaches as well as combined methodologies.

Table 9. Experimental and numerical research on gaseous film cooling techniques for rocket propulsion chambers.

	Period	Key Area of Investigation
Experimental Investigations		
Marek and Tacina [40]	1975	Influence of free-stream flow Turbulence
Black and Cuffel [78]	1976	Effect of wall cooling on the free stream flow dynamics
Gau et al. [76]	1991	Influence of swirl in the mainstream
Juhany et al. [70]	1994	Influence of coolant Mach number
Arrington et al. [45]	1996	Evaluation of cooling effectiveness in conical and bell nozzles
Kanda et al. [56]	1996	Influence of the external shock wave
Arnold et al. [69]	2009	Influence of the circumferential injector configuration
Arnold et al. [21]	2009	Influence of tangential slot injection
Shine et al. [44]	2012	Role of coolant injector design for multiphase film coolants
Hombsch and Olivier [42]	2013	Variable injection angles, coolant mass flux, and free stream properties
Numerical Investigations		
Martelli et al. [47]	2009	Practicality in dual bell nozzles
Peng and Jiang [75]	2009	Influence of the oblique shock wave
Shine et al. [43]	2013	Role of Coolant Injector Design in Multiphase Film Cooling

Table 10. Research on gaseous film cooling techniques for rocket propulsion chambers involving combined approaches.

	Period	Key Area of Investigation
Experimental and numerical investigations		
Aupoix et al. [73]	1998	High-speed film injection
Heufer and Olivier [68]	2008	Coolant injection in a laminar supersonic flows flow
Shine et al. [44]	2012	Influence of straight cylindrical holes on film cooling
Experimental and theoretical investigations		
Spalding [41]	1965	Correlation for film cooling performance
Stoll and Straub [22]	1998	Evaluation of thermal transfer in a film-cooled nozzle wall
Dellimore et al. [127]	2009	Correlation for a high-velocity hot gas stream
Experimental and analytical investigations		
O Connor and Sheikh [71]	1992	Influence of primary flow velocity
Matesanz et al. [48]	1993	Practicality of LES techniques
Kuo et al. [72]	1996	Influence of mainstream flow velocity
Analytical investigations		
Sellers [128]	1963	Correlation for multi-slot

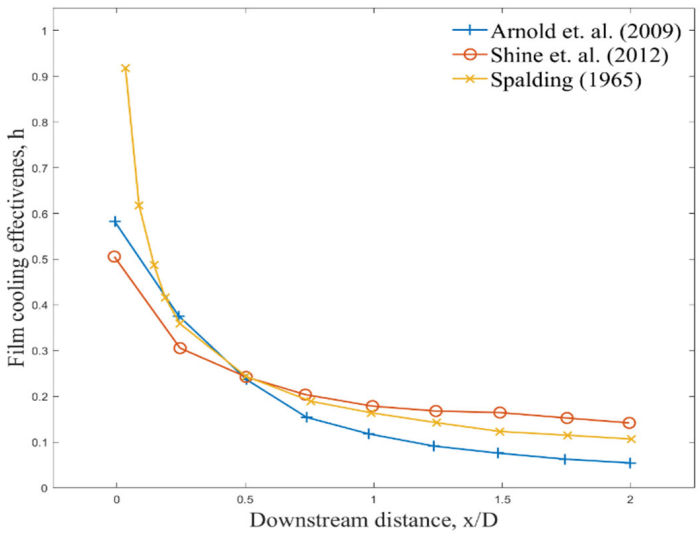

Figure 5. A comparison of adiabatic effectiveness predictions across various studies (reproduced from [9,21,41,44]).

4. Film Cooling for Gas Turbine Blades

The demand for enhanced thermal performance in gas turbine engines requires elevated operating temperatures, aligning with the requirements of modern turbine manufacturing to achieve effective and reliable power output. Therefore, this high inlet temperature

subjects turbine blades to excessive thermal stresses [9]. Increasing the turbine inlet temperature (TIT) is widely recognized as a key method for improving thermal efficiency and thermodynamic performance [129,130]. Consequently, when maintaining a constant pressure ratio, the net energy output increases, accompanied by an improvement in the rotor blades' performance, as explained by thermophysics principles.

Section 4.1 reviews various methods to address key challenges in gas turbine blade film cooling technology, including the phenomenon of jets in crossflow. Section 4.2 focuses on the mechanism of vortices. Figure 6a presents the 3D vortical structures observed experimentally during a film cooling analysis involving imperfect holes. Figure 6b depicts the hole exit, associated nomenclature, and the reference frame. Similarly, Figure 7 illustrates the most recent advancements in turbine inlet temperature from 1950 to 2010 as presented in Section 4.3. Finally, the effects of flow control parameters on film cooling performance are reviewed in Section 4.4.

4.1. Studies on Turbine Blades Film Cooling

Research on film cooling techniques for turbine blades has evolved significantly over the years, driven by the need to enhance turbine performance and durability under extreme thermal conditions. Early studies focused on basic cooling methods, including surface cooling methods and heat shields. However, as engine temperatures increased, more sophisticated approaches were explored. Consequently, the introduction of film cooling, which involves applying a coolant film or cooler air over the blade surface to protect it from high temperatures, became a key focus of research. This review uncovers that initial research on film cooling technology employed in gas turbine blades was experimental, with significant contributions from wind tunnel testing and empirical models. However, as CFD advanced, it enabled more accurate simulations of cooling flows, facilitating the optimization of cooling slot configurations, injection angles, and material characteristics. These studies have gradually shaped modern film cooling strategies, enhancing the efficiency and longevity of turbine blades in high-performance aerospace and power generation applications. The three-dimensional visualization of a flow field and the formation of a cross-flow jet are directly applicable to understanding and improving film cooling techniques for turbine blades, particularly in optimizing cooling flows and enhancing thermal protection. Gas turbines are employed in aerospace propulsion systems, fixed-site power generation, and numerous industrial applications, including trains, marine vessels, gas pipeline compressor drivers, and automobiles [9].

Developments in film cooling techniques applied to gas turbine blades are summarized below. One of the earliest challenges encountered by employing film cooling technology in turbine blade applications includes the phenomenon of jet in crossflow (JICF). Over the years, the jet-in-cross-flow phenomenon has been thoroughly investigated. Early research by Bergeles et al. [131] (1976) established foundational insights into jet-in-crossflow behavior through a detailed experimental investigation. The experimental study investigated the influence of the near-field flow region induced by a coolant jet introduced perpendicularly through a boundary into the primary flow. They reported that the effect of the blowing ratio on film cooling effectiveness was highly significant. Similarly, in 1982, Andreopoulos [132] investigated the role of jet-in-crossflow on cooling performance using a jet pipe. The investigation examined the velocity perturbation inside the pipe while varying the jet-pipe configuration parameters, including the crossflow velocity ratio. The findings demonstrated that the crossflow penetrated a considerable distance into the jet hole at low blowing ratios. Another investigation involving jet-in-crossflow was conducted by Andreopoulos and Rodi [133] in 1984. They employed an experimental method to study the turbulence characteristics in an adiabatic normal jet-in-crossflow, focusing on the influence of a jet

induced through a cylindrical opening into a transverse flow. The findings revealed that vortical dynamics were observed in both the wake and the shear transition zone. In 1994, Fric and Roshko [134] experimentally categorized different vortical structures in the wake of a transverse jet at high-velocity ratios. The investigation uncovered a significant finding regarding the origin of the vortical structures. It was discovered that this phenomenon did not originate from ejection from the jet but rather from vorticity in the crossflow wall boundary layer. Another experimental investigation focusing on vortical structures was conducted by Morton and Ibbetson [135] in 1996. They studied the mechanism behind the warping of these structures, including the role of hole configuration in their formation. The findings demonstrated that the hole configuration has a critical influence on the formation of the vortical structure. In 1997, Haven and Kurosaka [136] experimentally investigated the effect of vorticity on cooling effectiveness. The study examined the influence of hole exit designs on the near-field behavior of crossflow jets. They found that modifying the hole geometry controls jet lift-off and cross-flow entrainment.

This review highlights that numerous studies have provided insights into the intricate interaction between jets and cross-flow [137], covering aspects such as mixing, structures, and scaling [138], as well as numerical investigations of jets in cross-flow [139]. In a 2001 study by Lim et al. [140], a doughnut-shaped vortex was reported in the near-field region of the jet. Contrary to earlier reports, this phenomenon was not caused by the folding of ring vortices. However, it was observed that when the Jet interacts with the cross-flow, the doughnut-shaped vortex disrupts, generating windward and lee vortices. The influence of vortex-inducing jets on film cooling effectiveness depends on the nature of the vortices formed downstream in the flow field. Hence, to address this challenge, Johnston et al. [141] experimentally examined the effect of coolant hole inlet geometry on the near-wall flow field in 2002. They reported that the coolant hole inlet geometry has a significant influence on the near-wall flow field, although its effect on the far field is minimal. Studies have also investigated how an imperfectly manufactured hole can impact vortex formation in the cross-flow, influencing film cooling performance. In 2006, Jovanovic [142] experimentally investigated the role of a localized imperfection within a short perpendicular hole. The stability of the flow regime was reported in the absence of imperfection, while clockwise vortices were observed downstream. The findings revealed the formation of a vortex in the imperfect hole design, suggesting that this vortex has a significant influence on film cooling performance. In 2020, a study provided an overview of investigations into progress on optimization strategies for internal cooling and thermal flux rate analysis [9]. The study focused on factors such as blade tip leakage and the application of different cooling methods. It was reported that incorporating dimple surfaces in blade tip design enhanced the retention of optimal pressure differentials due to their ability to generate a pressure drop through the blade passage. It has been observed that increasing the TIT improves turbine performance but requires advanced cooling techniques to manage thermal stresses. Therefore, in 2014, a stator cascade was used to numerically investigate the effect of coolant film injection on cooling effectiveness [129]. Additionally, an experimental investigation was conducted to validate the flow phenomena observed in the vane passage in a supersonic wind tunnel. The findings revealed that the RANS model used in the simulations effectively assessed the losses caused by film cooling with significant accuracy. It is worth mentioning that factors such as blowing conditions were used to validate the flow characteristics associated with shear layer dynamics. Similarly, in 2017, a method was developed to analyze the internal coolant flow in gas turbine blades with intricate cooling configurations, which was then applied to evaluate coolant pressure profiles and optimize coolant mass flux [130]. The computed wall pressure and temperature profiles at

the turbine blade's center section were validated against experimental results, showing a strong correlation.

Enhancing turbine blade cooling efficiency is limited by scaling challenges in key flow parameters. These parameters include the mass flux, velocity, and momentum flux ratios. These phenomena, driven by changes in fluid properties, resulted in a density variation ratio on the blade core-line and spanwise-averaged thermally insulated film cooling performance. Hence, to address these problems, Sinha et al. [143] investigated film cooling performance in 1991 by studying a row of inclined holes on turbine blades using cryogenic coolant introduced on a flat, thermally insulated test plate. The findings revealed a significant decrease in spanwise-averaged adiabatic film cooling performance. Incorporating plenum designs into turbine blade thermal protection system configurations has enhanced cooling performance in gas turbine engines. In 2002, Peterson and Plesniak [144] experimentally investigated the influence of the flow field in multi-jet impingement with crossflow, using a variety of short coolant jet hole plenum designs. It is worth noting that the technique employed in this investigation involved utilizing a plenum to create a blade cavity, with air directed through it into the holes. The findings revealed that the orientation of the plenum exit greatly affects the flow field and, consequently, the heat transfer. Similarly, parameters such as aspect ratio, velocity ratio, and Reynolds number also influence the cooling performance of turbine blades. Over the years, researchers have employed experimental and numerical methods to investigate the influence of these flow parameters on the far-field flow. In 2003, New et al. [145] experimentally studied the vortical structures induced by an elliptical jet in crossflow using a water flow facility and the laser-induced fluorescence (LIF) method. Their findings suggested that the aspect ratio affected film cooling effectiveness only in the near field, with diminishing influence in the far field, where jet geometry influenced the flow orientation.

4.2. Mechanism of Vortices

Vortices are central to the dynamics of film cooling. Therefore, effective control of these vortices through innovative designs and optimized operations enhances cooling performance, contributing to the durability of turbine components under extreme thermal conditions. Jet-in-cross-flow dynamics are essential for understanding the mixing mechanisms in turbine airfoil cooling. These dynamics reveal intricate interactions with the surrounding fluid. Film cooling hole designs, including shaped, angled, or diffused holes, are engineered to reduce the adverse effects of vortices, such as jet lift-off. Moreover, carefully optimized injection angles help minimize disturbances caused by vortices.

A three-dimensional representation of windward (WV) and lee (LV) vortices is depicted in Figure 6a. Investigations have shown that windward vortices distort under velocity gradients [146]. Hence, vortices enhance turbulent mixing between the coolant jet and the hot mainstream gasses. Although this can affect cooling efficiency in certain cases, excessive mixing may dilute the coolant, diminishing its protective role. However, the interaction of vortices with the turbine surface boundary layer plays a crucial role in determining heat transfer coefficients. Experimental methods, such as particle image velocimetry, flow visualization, and computational simulations like CFD, are extensively employed to investigate vortex dynamics and their influence on film cooling performance. Current research aims to optimize the advantages of vortical mixing while ensuring the stability of the cooling film.

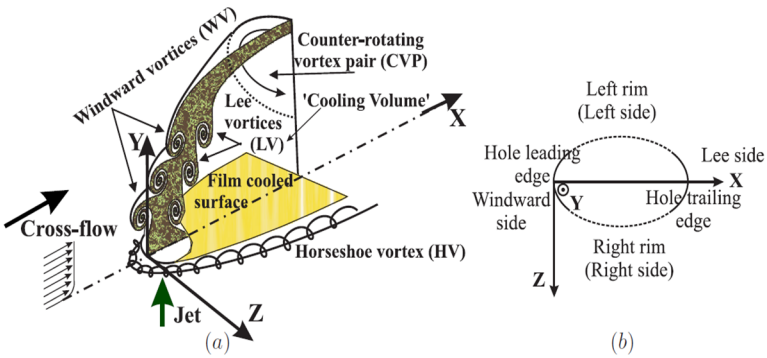

Figure 6. (**a**) Vortical structures resulting from the interaction of the jet cross-flow with experimentally observed film cooling properties. (**b**) The hole outlet, terminology, and coordinate framework (adapted from [146]).

4.3. Innovations in Gas Turbine Blade Film Cooling

In recent years, the turbine inlet temperatures of gas turbines have risen considerably and are projected to keep increasing [7]. This trend has been enabled by progress in materials, technology, and the adoption of advanced cooling methods for turbine blades. Introducing new materials and cooling technologies has driven the rapid increase in turbine firing temperatures and improved efficiency [147]. The first-stage blade is required to endure the harshest combination of temperature, stress, and environmental factors, and it is typically the most critical component in the machine [148]. Film cooling has been investigated in realistic and simplified geometries, utilizing either single or multiple holes. Conducting experiments on actual gas turbines is complex, and acquiring data under real engine conditions is highly challenging. As a result, there are limited studies involving real rotating blades.

The key parameters that characterize film cooling in turbine blade applications are the blowing ratio (*BR*), velocity ratio (*VR*), and momentum ratio (*MR*).

The blowing ratio is given by [146]:

$$BR = \frac{\rho_j \overline{u}_j}{\rho_\infty u_\infty} \tag{1}$$

where $\overline{u}_j$ and u_∞ denotes the surface average velocities of the jet and the free stream, respectively, while ρ_j and ρ_∞ represent the jet and free stream densities.

The velocity ratio is computed as [146]:

$$VR = \frac{\overline{u}_j}{u_\infty} \tag{2}$$

In the case of a unity density ratio, it is equal to the blowing ratio [146].

$$MR = \frac{\rho_j \overline{u}_j^2}{\rho_\infty u_\infty^2} \tag{3}$$

If the jet structure maintains uniform properties across the exit surface, the density ratio equals the product of the blowing and the velocity ratio. The investigation utilizing water channel experiments in a closed-return setup revealed that $BR = VR$, and $MR = VR^2$.

Thus, the jet cross-flow interaction is entirely determined by *VR*. In the case of compressible flow, the interaction is governed by a collective influence of all three factors [146] because of the varying densities.

In Figure 7, it can be seen that contemporary high-performance gas turbine engines generally function within a turbine inlet temperature (TIT) ranging from 1200 to 1500 °C [7].

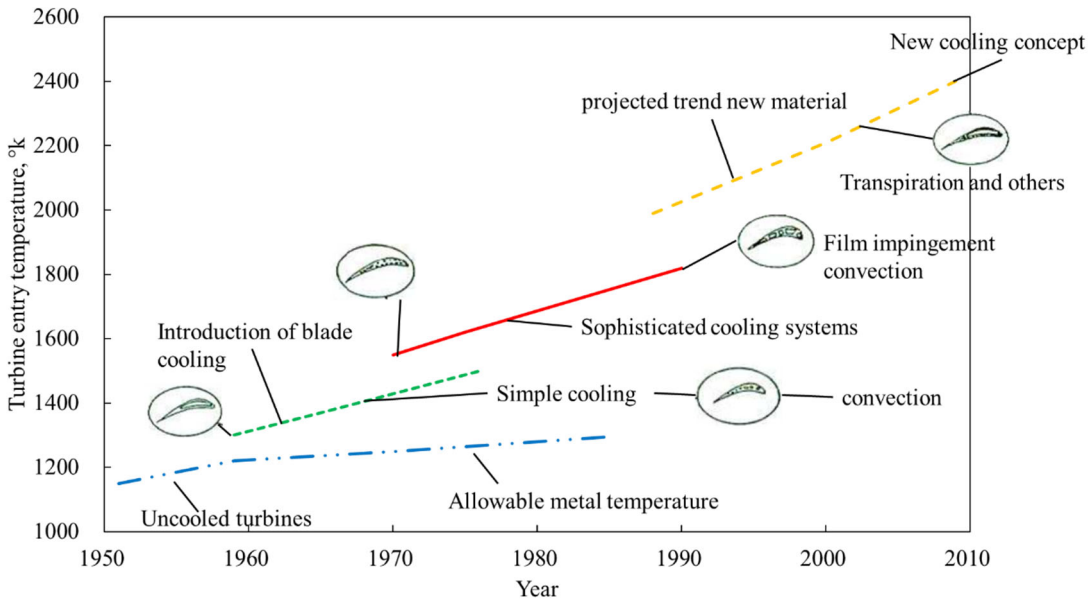

Figure 7. Trends in gas turbine inlet temperature in recent years (reproduced from [7]).

Investigations have shown that internal and external cooling techniques prevent turbine blades from exceeding melting temperatures. Figure 8 illustrates the airflow through the cooling inlet and the internal and external cooling channels, extending from the blade root to the airfoil sections.

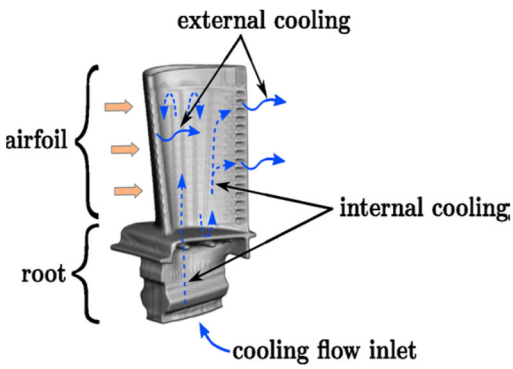

Figure 8. Internal serpentine channels within a turbine blade, revealed through transparent sections, show the cooling flow (adapted from [7]).

In external cooling, this bypassed air is released through precisely positioned narrow openings in the blade. Coolant air from within the blade is expelled through discrete holes or slots (Figure 9a), creating a thermal barrier that isolates the blade's outer surface from hot combustion gasses. The key benefit of this innovative technique is its ability to shield the turbine blades from the hot gasses exiting the combustion chamber. Film cooling holes in turbine blades are manufactured using electro-discharge, electrochemical, and laser drilling techniques [146]. In laser drilling, the material is melted and expelled from the hole. However, if the molten material is contained within the hole, it solidifies, causing a production defect (refer to Figure 9b,c). Film cooling techniques used in turbine components, such as high-pressure turbine vanes and blades, rely on five critical factors functioning simultaneously. These factors include internal convective cooling, external film cooling on the surface, material selection, thermal-mechanical design, and the pre-treatment of coolant

fluids [7]. This review has identified that a hybrid approach combining internal and film cooling is the most widely used method for cooling high-pressure turbine blades. The performance of a turbine blade film-cooling process is influenced by various physical properties, including the coolant-to-hot mainstream velocity ratio, blowing ratio, momentum ratio, pressure ratio, temperature ratio, density ratio, and turbulence intensity. Similarly, geometrical configurations have a significant influence on film cooling performance. As a result, the design of the turbine blade and the placement and distribution of film cooling holes have been extensively investigated.

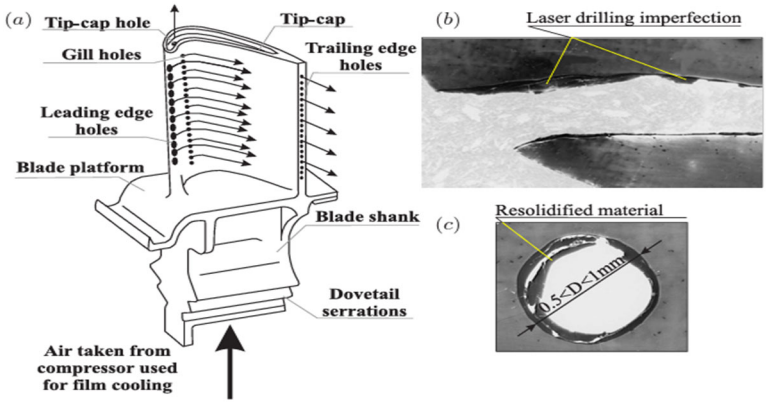

Figure 9. (**a**) A turbine blade with film cooling holes. (**b**) A side view of a laser-drilled hole showing imperfections. (**c**) A top view illustrating resolidified melt ejection from the film cooling hole (adapted from [146]).

4.4. Effects of Flow Control Parameters on Cooling Effectiveness

Optimal cooling effectiveness requires balancing key flow control parameters, such as the blowing ratio, streamwise angle, and hole geometry. A combination of CFD simulations with experimental testing is widely used to determine the most effective configurations for various operating conditions in gas turbine systems.

The progress on the effects of the blowing ratio, streamwise angle, and hole geometry on film cooling performance is reviewed and summarized below. In 1977, Pedersen et al. [79] investigated the effect of *BR* on film cooling effectiveness using a wind tunnel. Their findings demonstrated that the influence of compressibility was increasingly significant at higher *BRs*. Dring et al. [149] conducted a pioneering experimental investigation into the effect of *BR* on film cooling performance for a rotating blade in 1980. They employed different values for the coolant-to-mainstream mass flux in each blowing configuration. The experimental findings suggested that for each blowing configuration investigated, neither the coolant *BR* nor the coolant-to-mainstream density ratio affected the coolant film paths. Similarly, in 1999, Drost and Bölcs [150] experimentally investigated film cooling performance and heat transfer characteristics in turbine nozzle guide vanes. The study employed varying *BRs* to examine the cooling effectiveness on the pressure side and suction surface. It was reported that a higher *BR* reduced cooling efficiency on the suction side of the airfoil when a single row of film cooling holes was used, but enhanced cooling with a staggered hole configuration. In 2003, Mayhew et al. [151] studied the effect of low and high turbulence intensity on flat plate film cooling performance. The study measured adiabatic film cooling performance using the liquid crystal thermography method. Their findings showed that free stream turbulence influenced cooling, but its effect was minor and negligible at higher *BRs*. In 2012, Abdelghany et al. [147] employed the CFD software FLUENT® to investigate the role of film cooling in turbines, primarily to mitigate high temperatures. They performed a parametric analysis to assess the influence of blowing ratio, streamwise angle, and hole geometry on film cooling performance. The

study evaluated three blowing ratios (0.45, 1, and 2), three streamwise angles (30°, 60°, and 90°), and the impact of varying lateral and forward diffusion angles (ranging from 0° to 15°) on the cooling performance, while particularly emphasizing film cooling effectiveness on an adiabatic flat plate. The study concluded that the maximum centerline effectiveness of the blowing ratio occurs at a streamwise angle of 30° on the flat plate surface. For a row of 30° jets, the optimal blowing ratio is about 0.6, while for a row of 60° jets, it is slightly lower (around 0.5), and for a row of 90° jets, it is even lower (approximately 0.4). The highest lateral cooling effectiveness is observed in the LFDSA-15-15 configuration, which incorporates a fan-shaped diffuser with 15° divergence angles on both lateral sides across all blowing conditions.

In 1992, Hay and Spencer [152] conducted experimental studies to investigate the influence of streamwise orientation on the discharge coefficient. The study focused on the radius and chamfer configurations of the coolant holes and utilized the technique of determining the ratio of actual mass flow to ideal mass flow. Meanwhile, in 2000, Hale et al. [153] conducted a study on inner hole geometry, focusing on the role of a plenum in film cooling. Their findings showed that the geometry of the plenum significantly influenced film cooling performance, depending on both the injection hole length and the streamwise hole angle. Similarly, in 2001, Gritsch et al. [154] conducted an experimental investigation on the effects of the discharge coefficient using cylindrical coolant holes with a range of orientation angles. The results revealed that varying the orientation angle significantly influenced the losses observed at the coolant hole inlet section compared to the outlet section. However, the approaching boundary layer did not affect cooling beneath the film [155]. In 2003, Yuen and Martinez-Botas [156] used liquid crystal thermography to measure adiabatic effectiveness. Their study explored the influence of flow control parameters, such as the blowing ratio and streamwise angle, on the cooling performance. The results revealed that a 30° orientation angle exhibited the most effective cooling characteristics. Moreover, Yuen and Martinez-Botas [157] explored the influence of flow control parameters on the heat transfer coefficient by employing single jets with varying streamwise orientation angles. In 2006, Rutledge et al. [158] investigated the effect of leading and trailing edge roughness on heat transfer using a row of cooling holes on the suction side of a turbine vane. They observed that enhancing mixing led to a heat transfer coefficient twice as high as that of a smooth blade. Additionally, the relative effectiveness of film cooling showed improvement compared to an uncooled scenario.

In 1974, Goldstein et al. [159] experimentally investigated the influence of coolant hole configurations and flow characteristics on film cooling performance using film coolant introduced through discrete holes. Experimental results demonstrated that at increased *BRs*, the shaped hole configuration demonstrated significantly better effectiveness. In 1982, Goldstein and Yoshida [160] investigated how a laminar or turbulent boundary layer influenced a laminar or turbulent jet introduced through cylindrical inclined holes into a crossflow. The study examined a row of single circular injection holes with a defined lateral spacing between them. The results showed that turbulent jets achieved higher cooling effectiveness than laminar jets at the same *BR*. In 1990, Ammari et al. [161] experimentally evaluated the impact of the density ratio on heat transfer coefficients using expanded polymer surfaces and laser-based interferometry techniques. The findings revealed that experiments involving density ratios of unity must be scaled to achieve significant cooling efficiency. Similarly, in 1990, Pietrzyk et al. [162] investigated the influence of the density ratio on the flow distribution involving a row of inclined jets using hydrodynamics. The inclined jets were introduced into a crossflow with a known density ratio between the injectant and the free stream. The results showed that although variations in density ratio do not affect the flow patterns, they cannot be directly scaled

using either the blowing or momentum ratio. In 1991, Sinha et al. [143] studied the impact of the density ratio on film cooling performance involving a row of inclined holes using a conduction-error-free thermocouple setup. They observed that lowering the density and increasing the momentum ratio decreased the lateral average effectiveness. In 1996, Schmidt et al. [163] conducted a complementary investigation into the influence of compound-angle coolant hole configurations on adiabatic cooling performance using an experimental approach. The results suggested that compound-angle injection holes do not provide higher adiabatic effectiveness at the ideal momentum flux ratio; rather, they enhance cooling performance over a significantly extended range of momentum flux ratios. In 1998, Gritsch et al. [164] experimentally investigated the influence of hole configurations on the thermal transfer coefficient and film cooling performance using a variety of cooling hole geometries. Their analysis focused on a 2D thermal transfer coefficient profiling technique. The results showed that coolant holes with enlarged outlets exhibited lower thermal transfer coefficients at high *BRs*, unlike cylindrical hole configurations.

It has been observed that numerous investigations have focused on the influence of coolant geometries on the flow field [165], as well as how compound-angle film cooling holes influence thermal transfer coefficients [166]. In 1999, Lutum and Johnson [167] experimentally investigated the role of the aspect ratio in the film cooling process. Their findings indicated that an aspect ratio exceeding 7 has a negligible influence on film cooling effectiveness. However, it has been observed that shorter holes reduce the aspect ratio and tend to lower film cooling performance, while the shortest hole exhibited minimal cooling effectiveness [146]. In 2011, Chen et al. [168] experimentally investigated the influence of ramp angle and *BR* on film cooling efficiency. Their investigation examined a variety of ramp angles under varying blowing conditions. It was observed that incorporating a high ramp angle with a high *BR* significantly improved film cooling effectiveness. Investigations have shown that the position where the coolant is injected relative to the rear edge of the ramp is also a key factor affecting film cooling performance [169]. Typically, placing the ramp just before the holes greatly improves the adiabatic film cooling effectiveness. The formation of an additional pair of anti-kidney vortices occurs due to the mixed flow detaching from the horns of the sand-dune-shaped ramp. As a result, this ramp design significantly enhances film cooling across the entire surface [170]. In 2014, Davidson et al. [171] investigated the effects of different film-cooling hole geometries on the cooling performance of a thermally conductive turbine vane equipped with thermal barrier coatings (TBCs). Their research concluded that incorporating a TBC on the surface of an internally cooled vane maintained a nearly consistent cooling performance, even with substantial variations in the blowing ratio. The interaction of TBC and different film cooling configurations, aimed at providing a detailed evaluation of the thermal protection for a first-stage turbine vane, has also been examined. The findings revealed that applying TBC to the vane surface significantly improved the cooling effectiveness downstream of the coolant holes, even without active film cooling. Meanwhile, the presence of the TBC significantly decreased the fluctuations in overall cooling effectiveness caused by changes in blowing ratio, which in turn had a less negative impact on coolant jet separation. It was also found that standard round holes with TBC demonstrated comparable, if not superior, performance to round holes in a shallow transverse trench [172]. Investigations have revealed that the laterally averaged adiabatic film cooling effectiveness could be two to three times greater with an upstream ramp compared to the baseline scenario [173]. Although an upstream ramp reduces flow losses inside the film cooling hole, it increases the total aerodynamic loss in the film cooling system due to flow separation in the oncoming boundary layer [174,175]. The angle of the ramp is a crucial factor influencing film cooling performance. A parametric

study of these flow control parameters provided insights into their impact on film cooling effectiveness, particularly for turbine thermal protection applications [147].

Studies have demonstrated interdependency in parameter effects on system performance. For instance, shaped holes can help offset the drawbacks of high blowing ratios. However, a poorly chosen streamwise angle may diminish the advantages of advanced hole geometries. Achieving greater cooling effectiveness often involves trade-offs, such as higher coolant usage, increased aerodynamic losses, or greater manufacturing complexity. This review found that investigations on the effects of flow control parameters have primarily focused on experimental methods and are entirely applicable to gas turbines and combustion chambers. Therefore, further studies should employ a numerical approach, specifically the SPH method, owing to its advantages over other CFD techniques, with a focus on applications in thermal protection systems for hypersonic vehicles. A summary of studies on film cooling techniques for gas turbine blades is presented in Table 11.

Table 11. Research on film cooling for gas turbine blades.

	Period	Key Area of Investigation
Experimental Investigations		
Bergeles et al. [131]	1976	Jet in Cross Flow
Andreopoulos [132]	1982	Jet in Cross Flow using a jet pipe
Andreopoulos and Rodi [133]	1984	Jet in Cross Flow
Fric and Roshko [134]	1994	Vortical structures
Morton and Ibbetson [135]	1996	Vortical structures, The role of hole configuration
Haven and Kurosaka [136]	1997	Vorticity on cooling effectiveness
Moussa et al. [137]	1977	Jet in Cross Flow, Near Field Mixing
Smith and Mungal [138]	1998	Jet in Cross Flow, Mixing, structure, and scaling
Numerical Investigations		
Yuan et al. [139]	1999	Jet in Cross Flow
Montis et al. [129]	2014	Cooling losses
Abdelghany et al. [147]	2012	The role of film cooling in turbines
Zhang et al. [170]	2018	Film cooling enhancement, Upstream sand-dune-shaped ramp

5. Film Cooling for Hypersonic Vehicles

Aerospace vehicles flying at hypersonic speeds (above five times the speed of sound) must survive intense flight conditions [176]. Nevertheless, severe aero-thermal conditions pose major challenges for the materials and structures of the vehicle. The extreme heating experienced by hypersonic vehicles generates heat fluxes 3–7 orders of magnitude more than solar radiation, thermal gradients spanning thousands of degrees per centimeter, stagnation pressures reaching 10^7 Pascals, and ionized gas plasma that exacerbates material degradation [3,14,27]. Therefore, hypersonic vehicles require innovative injection methods, advanced coolants, and precise engineering designs for effective cooling under extreme flight conditions. As technology advances, film cooling remains a critical thermal protection

method for developing hypersonic aerospace vehicles. The challenges associated with hypersonic film cooling technology must be addressed through the materials used in key subsystems of hypersonic vehicles, such as the primary structure, leading edges, control surfaces, thermal protection systems, propulsion units, and guidance systems.

This section reviews emerging cooling technologies for hypersonic systems, including advancements in film and transpiration cooling methods and the development of new coolants. These technologies are critical for addressing the extreme thermal loads encountered during hypersonic flight. Progress in hypersonic film cooling technology from the late 2000s to the 2020s is reviewed in Section 5.1. Similarly, Section 5.2 illustrates the advancements in film-cooling research for hypersonic and high-speed aerospace applications over recent decades, as shown in Table 12. Section 5.3 reviews the development of transpiration coolant materials. Finally, transpiration cooling techniques for hypersonic applications are reviewed and summarized in Section 5.4.

5.1. Studies on Hypersonic Film Cooling

The progress in hypersonic film cooling technology from the late 2000s through the 2010s and into the 2020s, covering a variety of investigation methods, is reviewed and summarized below. Yang et al. [177] provided an overview of recent advancements in extreme-temperature materials in hypersonic conditions in their 2008 study. The elements investigated include oxidation resistance and the mechanical and physical characterization of thermal protection materials and their structures. These elements are vital for the safety of hypersonic vehicles and play a significant role in their design and manufacturing processes. Materials such as metallic thermal protection systems, ceramics, and carbon/carbon composites are commonly used. In 2010, Zhang et al. [178] created an integrated environment to perform multidisciplinary design optimization analyses under uncertain conditions. They incorporated trajectory analysis, aerothermal and aerodynamic calculations, thermal modeling, and structural simulations to optimize the thermal protection system (TPS) design for a Two-Stage-to-Orbit (TSTO) upper-stage vehicle. The results of the multidisciplinary optimization were evaluated with and without considering uncertainty, and a comparison was drawn between deterministic and probabilistic solutions. In 2016, Esser et al. [179] concentrated on refining a passive TPS by advancing the current thermal management strategies based on local heating conditions. These approaches are employed clearly and directly to establish the key parameters of thermal protection, leading to the formation of typical blunted geometries. Similarly, in 2016, Wu et al. [180] introduced a multidisciplinary design optimization method for thermal management systems, demonstrating how various fields are combined to improve cooling strategies in hypersonic applications. In 2017, Zhou et al. [8] developed an analytical method to address the heat transfer challenges in the TPS of hypersonic vehicles. They employed the Laplace transform and integral methods to model the thermal profile in the TPS under aerodynamic heating during flight. The findings were then compared to finite element analysis results, demonstrating strong agreement. In 2017, Kumar and Mahulikar [181] conducted a study using inverse heat conduction analysis with the Levenberg–Marquardt method to determine the aerothermal heating and thermal response of materials in a reusable hypersonic vehicle (RHV) using temperature data from within the TPS. They also examined how measurement errors affected the accuracy of the inverse results. The findings indicated that the Levenberg–Marquardt method is a robust technique for designing a lightweight passive TPS for an RHV through the inverse approach. In 2019, Yuan et al. [182] presented experimental evidence of performance improvements achieved through liquid film cooling in challenging conditions linked to hypersonic flight, highlighting its role in enhancing overall vehicle efficiency. They confirmed that liquid film cooling is viable for hypersonic applications and demonstrated that it pre-

serves aerodynamic efficiency due to the minimal impact of coolant injection on the primary airflow. Motivated by these findings, the study further investigated the characteristics of liquid film flow and the factors influencing it. Similarly, in 2019, Gou et al. [183] devised a thermal management strategy for hypersonic vehicles. Their approach combined a passive thermal protection system (TPS) with an active cooling network (ACN) using kerosene as the coolant. The results demonstrated that enhancing the active cooling system reduces passive TPS weight while increasing the coolant mass flow rate. In 2020, Zhang et al. [184] provided an overview of the advancements in active thermal protection techniques for air-breathing hypersonic vehicles over the past several decades. Their review extensively addressed the core challenges and recent advancements in regenerative, film, transpiration, and combined cooling strategies. In 2021, Gou et al. [185] provided important insights into thermal management technologies for dissipating aerodynamic heat in a hypersonic cruiser. They achieved this with a passive TPS, transport through a regenerative cooling (RC) network, and reuse using the RC network and a thermoelectric (TE) conversion component. They also outlined a design roadmap for the thermal management system and explained the impact of the overall equivalent heat transfer coefficient. The findings indicated that as this coefficient increases, a greater proportion of aerodynamic heat is managed, transported, or reused, while the size of the passive TPS decreases. In 2023, Karimas and Salonitis [186] conducted a comprehensive literature review of oxide-based and non-oxide ceramic matrix composites (CMCs). They analyzed the mechanical and thermal properties, including their classification and production methods. Furthermore, existing manufacturing processes were systematically reviewed and comparatively analyzed. The review also proposed a research and development roadmap to advance these materials, promoting broader use in aero-engine applications. Similarly, in 2023, Guo et al. [187] thoroughly explored the challenges of thermal management and power generation for scramjet engines in hypersonic vehicles operating in high-temperature environments, including issues like pneumatic heating and combustion effects. The newly developed Power and Thermal Management System (PTMS) effectively cools hypersonic vehicles traveling at Mach 6–7 while preventing coking and damage within the scramjet cooling channels. Furthermore, it lessens the dependency on fuel as a coolant, with the added weight compensated by increased endurance during extended hypersonic flights. Also, in 2023, Coutinho et al. [188] analyzed the challenges of developing lightweight thermal management systems for all-electric and hybrid-electric aircraft, which face higher heat loads than traditional systems. These challenges are tied to both environmental and technological aspects in propulsion electrification. They reported that liquid cooling loops integrated with ram-air heat exchangers resulted in the most effective thermal management solution.

The materials currently used in the design and manufacturing of hypersonic vehicles have limitations in terms of structural durability under extreme temperature conditions. Therefore, in their 2024 investigation, Peters et al. [176] examined the need to develop new materials, from theoretical research to large-scale production. Similarly, in 2024, Lei et al. [189] explored the role of cutting-edge cooling concepts on sophisticated hypersonic vehicles and their potential impact on future aerospace technology. They proposed a novel cooling approach that combines spike and transpiration techniques to address the thermal protection challenges of hypersonic vehicle nose cones. They found that increasing the coolant mass flow surface temperature along the nose cone for the combined setup decreases the temperature reduction rate. Under certain incoming flow conditions, selecting the correct coolant mass flow is crucial for achieving the best thermal protection. Also, in 2024, Lee et al. [190] reviewed advancements in thermal protection systems, detailing and contrasting the distinctive phenomena of supersonic regimes with those of low-speed flows. It was observed that these characteristics posed considerable challenges due to

factors such as drag, noise, and heat generation. Their analysis encompassed critical flow phenomena, including boundary layer transition mechanisms, shock wave dynamics, and the formation of sonic boom. In 2024, Jung and Ko [191] investigated how integrating AI with radiative cooling technology advances the field by tackling difficulties associated with traditional methods and providing strategic solutions to worldwide challenges. Similarly, the impact of artificial intelligence and machine learning on optimizing cooling technologies for hypersonic vehicles highlights a novel approach that combines elements of computer science, engineering, and aerodynamics.

Currently available materials for thermal protection systems (TPSs), which must operate under extreme thermal stresses, impose limitations on hypersonic applications. Therefore, this review identifies the key challenges and research gaps associated with currently available materials and outlines them as follows:

(1) Investigations have shown that extreme-temperature materials, such as ultrahigh-temperature ceramics, fail to meet the requirements of advanced hypersonic vehicles. Therefore, further studies should focus on the oxidation reaction process and microstructural engineering;
(2) Studies on ultrahigh-temperature ceramics should also aim to develop methods for improving strength, toughness, and thermal shock resistance;
(3) Further studies should aim to develop techniques to test advanced TPS materials under hypersonic operating conditions, emphasizing numerical methods;
(4) Future research should prioritize systemically designed, triple-resistant (oxidation/load/thermal-crack) coatings for thermal protection, ensuring suitability for hypersonic vehicle applications;
(5) Existing research highlights a lack of thorough investigation into how high-speed and hypersonic flows affect transpiration cooling performance. Therefore, future research should investigate how mainstream flow parameters affect transpiration film cooling under hypersonic conditions.

5.2. Innovations in Hypersonic Film Cooling

Thermal protection systems for hypersonic vehicles focus on safeguarding the outer surface and the engine's internal components, as both experience severe thermal environments [184]. Hypersonic vehicles (Mach > 5) [192,193] are designed to achieve ultra-high speeds and are vital for the strategic progress of civil and military aerospace engineering. The air-breathing scramjet [192] is considered the most promising propulsion technology for these vehicles [184]. As the flight Mach number and cruise duration increase during hypersonic flight, the vehicle's outer surface and the internal components of the main powerplant are exposed to intense thermal conditions. Consequently, thermal protection technology is crucial for air-breathing hypersonic vehicles to survive these extreme thermal conditions [194]. These studies employed a range of methods, including experimental, numerical, analytical, and theoretical, used individually or in combination.

Table 12. Recent advances and innovations in film cooling techniques for hypersonic and high-speed aerospace applications.

Source	Year	Type of Study	Area of Investigation, Flow Type
Brown and Roshko [195]	1974	Experimental	Turbulent mixing, Supersonic
Goebel et al. [196]	1990	Experimental	Behavior of supersonic mixing layers, Supersonic flow

Table 12. *Cont.*

Source	Year	Type of Study	Area of Investigation, Flow Type
Kamath et al. [197]	1990	Experimental and Numerical	Effect of incident oblique shocks
Kanda et al. [198]	1991	Analytical	Cooling requirement on the engine characteristics
Buttsworth [199]	1996	Theoretical	Interaction of an oblique shock, Supersonic mixing
Glass et al. [200]	2001	Numerical	Convective and transpiration cooling and oxidation protection
Fan et al. [201]	2007	Experimental	Characterization of thermally cracked kerosene
Wang et al. [202]	2008	Numerical	Viscous mixing of the flow field, Supersonic
Willard et al. [203]	2009	Numerical	Scramjet/Ramjet Heat Exchanger Analysis
Soller et al. [204]	2009	Experimental	Cooling techniques (using both metallic and ceramic materials), Supersonic
G'enin and Menon [205]	2010	Numerical	Normal and oblique shock interactions, Turbulent flows
Hou et al. [206]	2012	Experimental	Heat Sink and Conversion of Catalyst
Zhang et al. [207]	2014	Numerical	Effects of Shock Waves, Supersonic
Hou et al. [208]	2014	Experimental	Cooling and coke deposition of Hydrocarbon
Zhang et al. [209]	2014	Numerical	Validation of a scramjet engine, Supersonic
Liu et al. [210]	2018	Numerical	Modeling of supercritical catalytic steam
Zuo et al. [124]	2018	Numerical	Evaluation of film and regenerative cooling, Supersonic
Jiang et al. [211]	2019	Experimental and numerical	Effect of shock waves on transpiration cooling, Supersonic

5.3. Development of Transpiration Coolant Materials

Recent advancements in transpiration coolants focus on improving coolant materials, refining delivery systems, and increasing energy efficiency to address the challenges of extreme conditions such as high-speed or hypersonic flight. These innovations include novel gaseous and hybrid coolants, adaptive systems for real-time thermal management, and integration with other cooling technologies for superior performance. Existing literature indicates that the coolant for transpiration cooling is generally a gas, with air being the most widely used due to its easy availability during flight and the fact that it usually does not require storage. However, collecting the air demands energy. Moreover, an additional cooling cycle system is demanded when the incoming air temperature is excessively high during high-performance flight [184]. Besides incoming air, other gasses such as nitrogen, argon, helium, carbon dioxide, and hydrogen can also serve as coolants [212,213]. The experimental setup for gas transpiration cooling [214] is illustrated in Figure 10. Similarly, Figure 11 demonstrates that helium offers excellent cooling performance in gas transpiration cooling [212]. Numerous studies utilize inert and non-oxidizing gasses as coolants because of their high specific heat capacity and oxidation resistance. However, for hypersonic vehicles powered by hydrocarbons, the weight of the coolant introduces a significant mass penalty that must be addressed. Liquid coolants typically produce a stronger phase change cooling effect than gas coolants. However, they encounter substantial resistance from viscosity and inertia within the porous medium, which places considerable demands on the booster servo mechanism in the coolant supply system. Currently, transpiration

cooling utilizes two main types of coolants: the engine's fuel and oxidant, as well as the supplemental coolant carried by the vehicle [184].

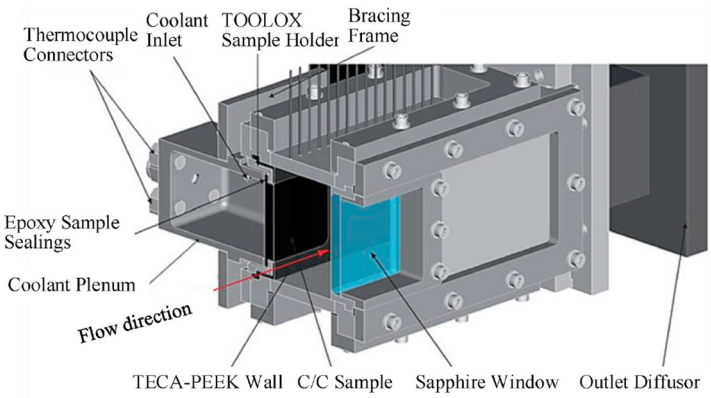

Figure 10. Setup of the testing apparatus for gas transpiration cooling (adapted with permission from [184,214,215], Elsevier, 2025).

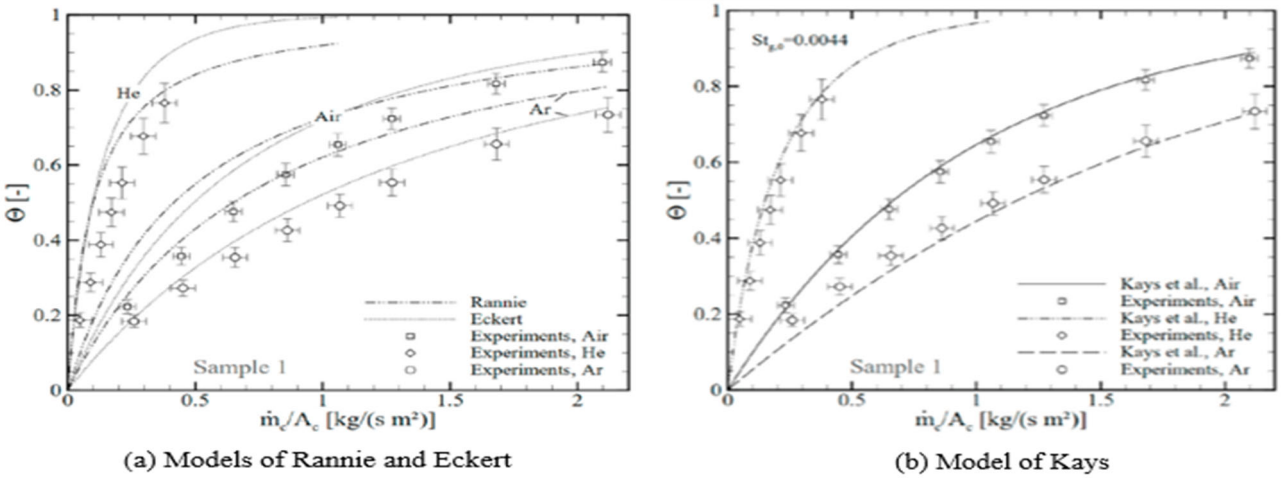

Figure 11. Evaluation of transpiration cooling using various coolants (adapted with permission from [184], Elsevier, 2025).

The first group of coolants primarily consists of liquid oxygen [216] liquid hydrogen [217], methane [218], and aviation kerosene [219,220]. The second group is mainly made of water [221]. Figure 12 depicts an example of water transpiration cooling in an arc wind tunnel. A clear phase interface forms within the porous media during the phase change process of transpiration cooling [222], where a well-defined two-phase region divides the saturated liquid phase from the saturated vapor phase [184]. Figure 13 illustrates a typical test rig for phase change transpiration cooling [222]. The phase transition process retains a large quantity of heat, enhancing cooling efficiency [223]. This phenomenon is due to the oxidant's poor stability, high oxidizing properties, and low latent heat of vaporization [216]. Hence, greater emphasis is placed on transpiration cooling using the vehicle's fuel [184].

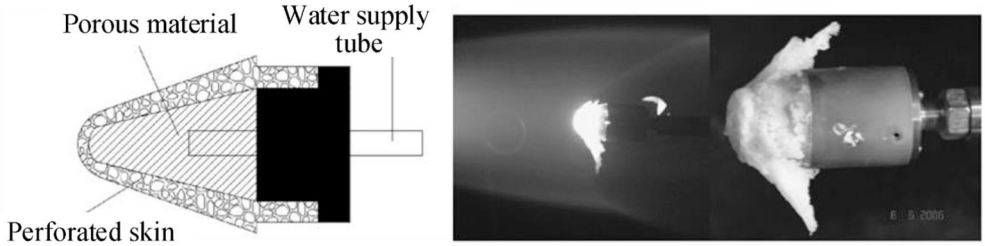

Figure 12. Water transpiration cooling experiment (adapted with permission from [184,221], Elsevier, 2025).

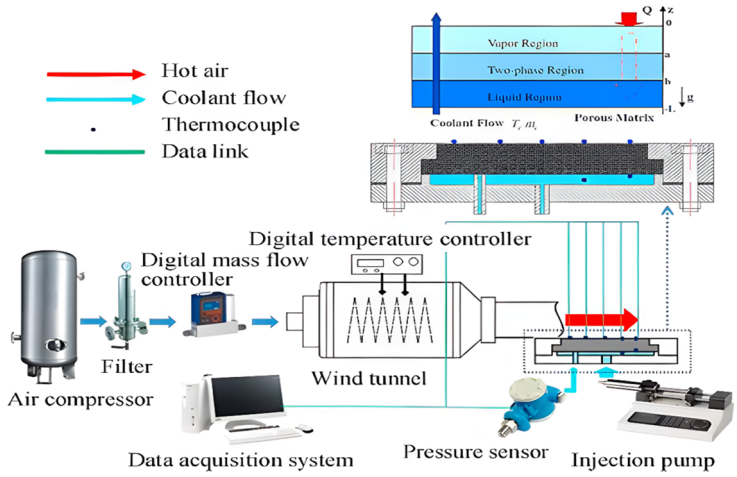

Figure 13. Phase change in porous media during transpiration cooling (adapted with permission from [184,222], Elsevier, 2025).

Another significant aspect of liquid transpiration cooling research is the critical heat flux, which seeks to keep the wall surface at the highest feasible temperature to reduce the amount of coolant required [223]. It has been demonstrated experimentally that water is highly efficient for cooling the porous cone [224]. It has also been shown that when the liquid transpiration coolant is a macromolecular hydrocarbon fuel, like aviation kerosene, the cracking and coking properties significantly impact transpiration cooling [219]. Meanwhile, the emergence of solid coolants dates back to the Cold War between the Soviet Union and the United States. During this time, the United States demanded high-temperature-resistant materials for vehicle thermal shielding and proposed the idea of a liquid metal transpiration coolant [225]. Placing a low-melting-point metal onto a porous media substrate with a high melting point and resistance to high temperatures, the metal melts and facilitates transpiration cooling in elevated temperature conditions, while a metal oxide layer forms on the porous surface [226]. Solid coolants generally possess very high latent heat and do not require a supplementary servo mechanism for free transpiration; however, their reusability is restricted, making them relatively expensive.

Experimental data indicate that using tungsten as the porous substrate, with copper as the coolant, to cool nozzle throat components yields superior cooling performance [225]. Another form of solid transpiration coolant is a hydrogel, which operates through free transpiration cooling [184]. Hydrogel has been used as a coolant in transpiration processes [227]. Porous media play a critical role in transpiration cooling by enabling uniform coolant distribution and efficient transport across the surface. The specific properties of the porous media, such as pore size and permeability, influence the coolant's flow behavior, phase change dynamics, and overall effectiveness in heat transfer. This makes the design and optimization of the porous medium crucial for achieving high-performance transpiration cooling systems.

5.4. Transpiration Cooling Techniques for Hypersonic Applications

Researchers have primarily focused on the effects of compressibility in supersonic airflow and shock waves on transpiration cooling for high-velocity applications, as the practical implementation of this cooling method remains neither systematic nor fully developed. The primary research approaches involve experimentation and numerical simulations, although theoretical and mechanistic investigations are still limited [184]. Northam et al. [228] verified that transpiration cooling performs exceptionally well under supersonic mainstream conditions, a finding further supported by the experimental investigations of Soller et al. [204] and the numerical analysis conducted by Glass et al. [200]. Studies have demonstrated that the shock wave in supersonic mainstream flow significantly reduces the effectiveness of transpiration cooling; however, employing materials with higher thermal conductivity, like bronze, can enhance this cooling performance [229]. Shock waves diminish the effectiveness of transpiration cooling, primarily because the rise in local pressure and temperature leads to a reduction in fluid velocity and an increase in the temperature of the porous media materials [211]. Studies have also indicated that enhancing the coolant mass flow rate can mitigate the negative impact of shock waves on local cooling [230]. Currently, research focuses primarily on two key aspects: one is the transpiration cooling of the strut on the aircraft engine, while the other investigates the coupled transpiration and regeneration cooling modes in the hypersonic vehicle, both of which occur in the supersonic mainstream [184], as shown in Figures 14 and 15. Investigations into the effect of transpiration cooling in the high-enthalpy supersonic mainstream have demonstrated that the transpiration cooling strut is efficient and can successfully isolate aerodynamic heating [231,232]. A study has determined that an optimal wedge angle for the strut exists, providing the most effective cooling for transpiration analysis [233]. It has been noted that transpiration and film cooling enhance the strut's cooling efficiency, resulting in a more homogeneous surface temperature [211]. Studies have examined the problem of thermal cracking and coking in porous media when hydrocarbon fuel is used as a coolant [219,220,234–237], as well as the interaction between flow in the cooling channel and transpiration on the wall [238]. However, it has been observed that there is a lack of research on the effects of the supersonic mainstream on transpiration cooling [184]. Integrated cooling technologies in hypersonic systems, including active/passive hybrid cooling, regenerative/film hybrid cooling, and regenerative/transpiration hybrid cooling, offer comprehensive thermal protection, boost system efficiency, prolong component life, and improve overall mission safety and performance, making them crucial for the effective operation of hypersonic vehicles.

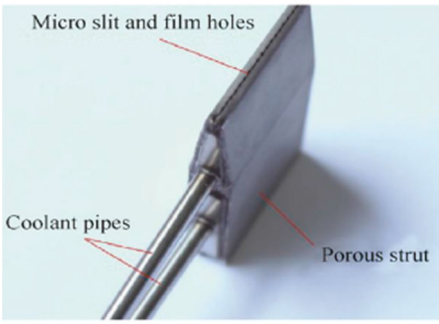

Figure 14. Porous strut featuring micro-perforations on the leading edge (adapted with permission from [184,239], Elsevier, 2025).

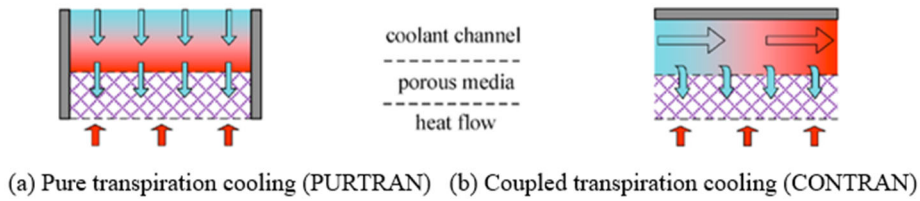

Figure 15. Categorization of transpiration cooling (adapted with permission from [184], Elsevier, 2025).

6. Perspectives on Future Film Cooling Research

Film cooling, a crucial approach for thermal management in gas turbines, combustion chambers, and hypersonic systems, is expected to prioritize several critical aspects to optimize performance and enhance operational capabilities. The areas of focus include: creating thermal barrier coatings and materials to improve heat resistance and durability; designing innovative cooling hole shapes and layouts to enhance heat dissipation and reduce coolant consumption; researching new coolant compositions with exceptional thermal properties and stability under extreme conditions; developing advanced predictive tools to optimize cooling efficiency and lower experimental costs; integrating film cooling with methods such as transpiration cooling to achieve synergistic benefits; and the development of smart cooling systems that adjust to dynamic thermal loads during flight. Research could focus on creating new materials with higher thermal conductivities and improved structural integrity for extreme conditions. These materials have the potential to enhance cooling efficiency and longevity. Enhanced cooling effectiveness requires the exploration of novel cooling hole shapes and configurations, such as shaped holes or micro-impressions. Advanced computational models will be crucial in designing these features. Employing 3D printing and other advanced manufacturing methods could enable more intricate cooling geometries, which are challenging or unfeasible to achieve with conventional techniques. This advancement can result in more effective cooling strategies.

Similarly, improved simulation tools that integrate fluid dynamics, heat transfer, and structural analysis might provide more precise predictions of cooling performance, aiding in the creation of more efficient designs. As environmental regulations become stricter, research may shift toward an eco-friendly approach, potentially incorporating systems designed to reduce emissions. Subsequently, integrating film cooling with active cooling techniques, such as phase-change materials or adjustable geometry hardware, could offer innovative methods to enhance cooling performance. These advancements address the challenges of increasing thermal loads in next-generation aerospace vehicles, ensuring safety, efficiency, and performance in extreme operating conditions.

7. Conclusions and Discussion

Film cooling technology is essential for the thermal control and efficiency of high-temperature components found in combustion chambers, gas turbine blades, and hypersonic vehicles. The core idea is to apply a thin layer (or "film") of cool air or another fluid onto the component's surface to protect it from excessive heat and minimize heat transfer. This review examines the fundamentals, applications, and techniques of film cooling. It highlights innovations in combustion chambers, gas turbine blades, and hypersonic vehicles and presents advances in efficient film cooling methods for these applications. The available literature confirms the effectiveness of gaseous film cooling techniques for combustion chambers, turbine airfoils, and liquid film cooling for hypersonic vehicles.

In combustion chambers, film cooling helps shield the walls from the intense temperatures produced during combustion. This is particularly critical for materials that cannot withstand the direct heat from the flames or exhaust gasses. Similarly, gas turbine

blades operate at very high temperatures, frequently surpassing the material tolerance of the blades. Therefore, film cooling is used to protect critical areas of the blade from thermal damage. On the other hand, hypersonic vehicles traveling at speeds above Mach 5 encounter intense aerodynamic heating as air friction increases surface temperatures to potentially destructive levels. Hence, film cooling aids in managing these heat loads on critical surfaces such as leading edges, fuselage, and engine components. In combustion chambers, cool air is introduced through holes or slots in the chamber walls to form a protective layer, preventing thermal degradation and enhancing the chamber's lifespan. In the case of turbine blades, coolant flow is introduced through small holes arranged in intricate patterns, forming a thin cooling layer on the surface and thereby reducing wall temperature. Similarly, film cooling is employed to inject coolant, typically air or other fluids, to create a protective layer on the surfaces of hypersonic vehicles. This cooling film shields against the intense heat generated by high-speed flight. Achieving uniform coverage, managing heat fluxes under varying conditions, and minimizing the impact on combustion efficiency are key challenges in combustion chambers. The key challenges in turbine blade film cooling include maintaining effective cooling without compromising turbine efficiency, ensuring coolant hole durability under high-pressure and high-temperature conditions, and managing complex flow dynamics. In hypersonic applications, achieving effective cooling at hypersonic speeds, where temperature gradients and heat fluxes are significantly more extreme, is a major challenge. Other key challenges include choosing the correct coolant, managing the interaction between the cooling film and the surrounding high-temperature airflow, and reducing cooling demands without compromising aerodynamic performance. The common challenges across all applications can be summarized as: achieving effective heat transfer reduction without compromising performance from higher cooling demands, minimizing coolant flow rate while ensuring the film remains durable and effective over time, managing the interaction between coolant flow and surrounding hot gasses to avoid issues like flow separation, vortex formation, and other harmful effects, and choosing suitable materials for the cooling system that can withstand both mechanical and thermal stresses.

The lifespan of vital components of combustion chambers can be extended and overall efficiency enhanced by optimizing design factors such as injection type, hole configuration, and coolant flow distribution. Gaseous film cooling has proven effective, demonstrating its viability as a cooling method through both experimental and numerical validation, especially for nozzle cooling applications. Existing correlations can predict film cooling effectiveness, supporting the design of rocket nozzle cooling systems. In gas turbine component cooling, particularly for high-pressure turbine vanes and blades, five key elements have been identified that must operate simultaneously. These elements include internal convective cooling, external film cooling, material selection, thermal-mechanical design, and the selection or treatment of coolants. Studies have proven that internal and external cooling techniques are employed simultaneously to prevent blade temperatures from exceeding their melting point.

The cross-flow jet in turbine blade film cooling performs several important roles: enhancing heat transfer, preventing hot spots, controlling boundary layer behavior, increasing cooling efficiency, and minimizing thermal fatigue. Turbine inlet temperatures have increased significantly and are expected to continue rising, driven by recent advancements in materials and the implementation of advanced cooling techniques for turbine blades. Advancements in materials and cooling technologies have enabled a rapid increase in turbine firing temperatures, leading to enhanced turbine efficiencies.

As technology advances, film cooling is expected to remain a vital component of thermal protection strategies for hypersonic aerospace vehicles. Cutting-edge injection

techniques, advanced coolants, and precise design facilitate efficient cooling even under extreme hypersonic flight conditions. Recent developments in transpiration coolants focus on enhancing coolant materials, optimizing delivery systems, and improving energy efficiency to tackle the challenges posed by extreme conditions such as high-speed or hypersonic flight. Tables 2–7 summarize significant studies on liquid film cooling in rocket thrust chambers. Key focus areas include film viability analysis, applications in the nozzle section, coolant types, heat transfer and evaporation analysis, flow phenomena and transpiration analysis, performance analysis, flow modeling, and related topics. A review of the different factors affecting the effectiveness of film cooling in these referenced applications is provided and summarized in Table 8. Similarly, various experimental and numerical studies on gaseous film cooling techniques in rocket propulsion chambers are reviewed and summarized in Table 9. Additionally, studies examining combined approaches are reviewed and summarized in Table 10.

The development of vortical structures, including CRVPs, WV, and LV vortices, around turbine airfoils causes the coolant jet to detach from the surface. This phenomenon decreases film cooling efficiency and increases exposure of turbine components to excessive temperatures. Effective management of these vortices through advanced design techniques and optimized operational approaches, such as incorporating shaped, angled, or diffused film cooling holes, can help minimize the detrimental effects of these vortical structures. On the other hand, the performance of parameters like blowing ratio (BR), streamwise angle, and hole geometry is interdependent. Shaped holes have been found to alleviate some of the disadvantages associated with high blowing ratios. However, selecting an inappropriate streamwise angle can diminish the benefits of advanced hole geometries. Enhancing cooling effectiveness typically requires balancing trade-offs, such as increased coolant consumption, higher aerodynamic losses, or more complex manufacturing processes.

This review identifies the following research challenges in film cooling for turbine components, rocket combustion chambers, airfoil surfaces exposed to combustion gasses, and hypersonic vehicles subjected to extreme thermal stress:

(1) Tangential injectors are commonly used, but the influence of other geometric factors has not been thoroughly analyzed. Therefore, further research is recommended to understand how coolant holes and injection angle configurations affect film cooling performance;
(2) Research has shown that coolant entrainment in the film leads to substantial coolant loss and a shorter liquid film-cooled length. Hence, additional investigations are required to quantify the impact of factors such as fluctuating mainstream flow, coolant hole or slot design, and their length;
(3) Despite extensive experimental research on liquid film cooling, the influence of injector orientation remains understudied. Future studies should numerically investigate how different coolant injector configurations affect film cooling performance;
(4) Aspects such as hole shape, hole spacing, and hole length have not been studied as thoroughly as in gas turbine film cooling applications. Therefore, future investigations of these aspects should focus on hypersonic film-cooling applications using the SPH method, given its advantages over conventional CFD approaches;
(5) It is crucial to understand that transient wall-side loads may cause structural damage to the surface of a hypersonic vehicle. There is a lack of scientific research on the impacts of these transient wall-side loads. Therefore, research in this area is vital for understanding the additional side loads induced by film coolant–freestream flow interactions;

(6) Hypersonic vehicles operate under supercritical conditions; therefore, further research is necessary for developing models that accurately predict the liquid film cooling length for hypersonic vehicle applications.

Progress in film cooling has led to better-cooling hole designs, the adoption of advanced materials, and advanced computational techniques to optimize cooling patterns. Additionally, there is growing interest in incorporating active cooling systems, such as adaptive or regenerative technologies, to improve cooling performance in these high-temperature environments.

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Data Availability Statement: The review presents original contributions in the field, identifies research gaps, and summarizes the authors' findings, including recommendations for further investigation. However, additional inquiries may be directed to the corresponding author.

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Nomenclature

AR	Aspect ratio
c	Heat capacity (unit: J/(kg · K))
h	Film cooling effectiveness
M, *BR*	Blowing ratio
MR	Momentum ratio
Tu	Turbulence in mainstream
VR	Velocity ratio
x/D	Downstream distance
u	Velocity (unit: m/s)
ρ	Density (unit: kg/m^3)

Subscript

(infinity)	free stream
j	jet
c	coolant

Abbreviation

CFD	Computational fluid dynamics
CIC	Coolant inter-cooling
CMC	Ceramic matrix composite
CPC	Coolant pre-cooling
CRVPs	Counter-rotating vortex pairs
DNS	Direct numerical simulation
LES	Large eddy simulation
LV	lee vortex
RANS	Reynolds-averaged Navier–Stokes
SPH	Smoothed Particle Hydrodynamics
SST	Shear stress transport
TBC	Thermal barrier coating
TEG	Turbine exhaust gas
TIT	Turbine inlet temperature
TPS	Thermal protection system
WV	Windward vortex

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